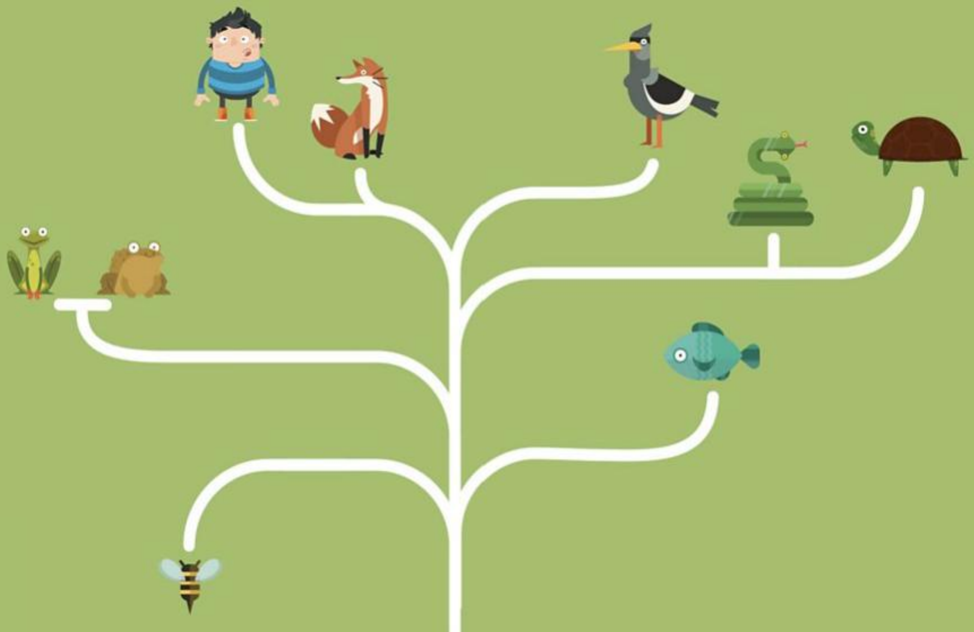
05

Decision Tree Classification



Classification

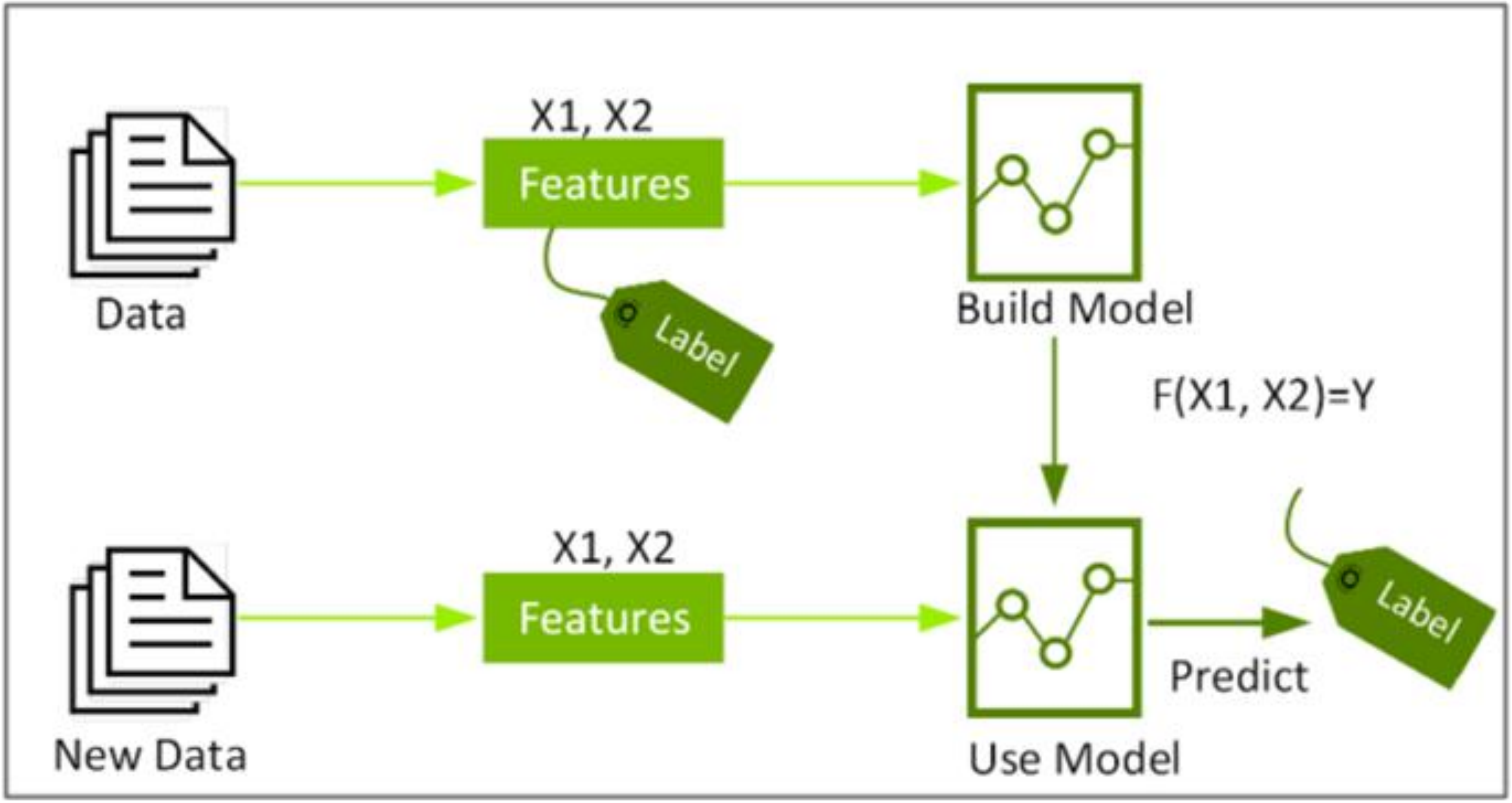
“

- กระบวนการสร้างโมเดลเพื่อจำแนกข้อมูลใหม่ ให้อยู่ในกลุ่มที่กำหนดมาให้จากกลุ่มตัวอย่างข้อมูลเดิม
- ข้อมูลที่ใช้ในการสร้างโมเดลเราจะเรียกว่า training data

Class / Label

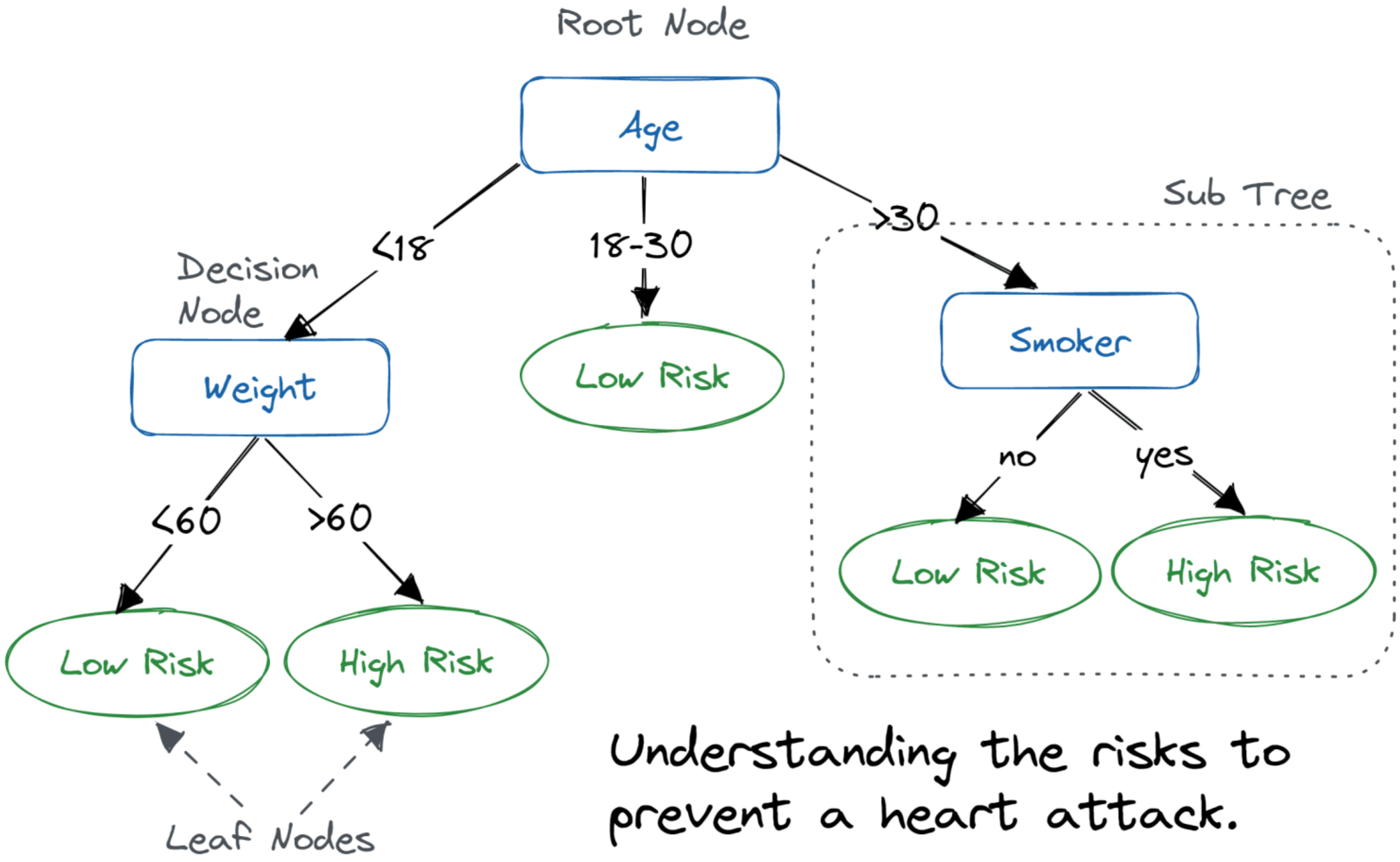
Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

Classification | Predictive Modeling



Classification Algorithms

- k-Nearest Neighbor (KNN)
- Decision Tree **



Why Decision Tree?

- Decision trees are powerful and popular tools for classification and prediction.
- Decision trees represent rules, which can be understood by humans and used in knowledge system such as database.

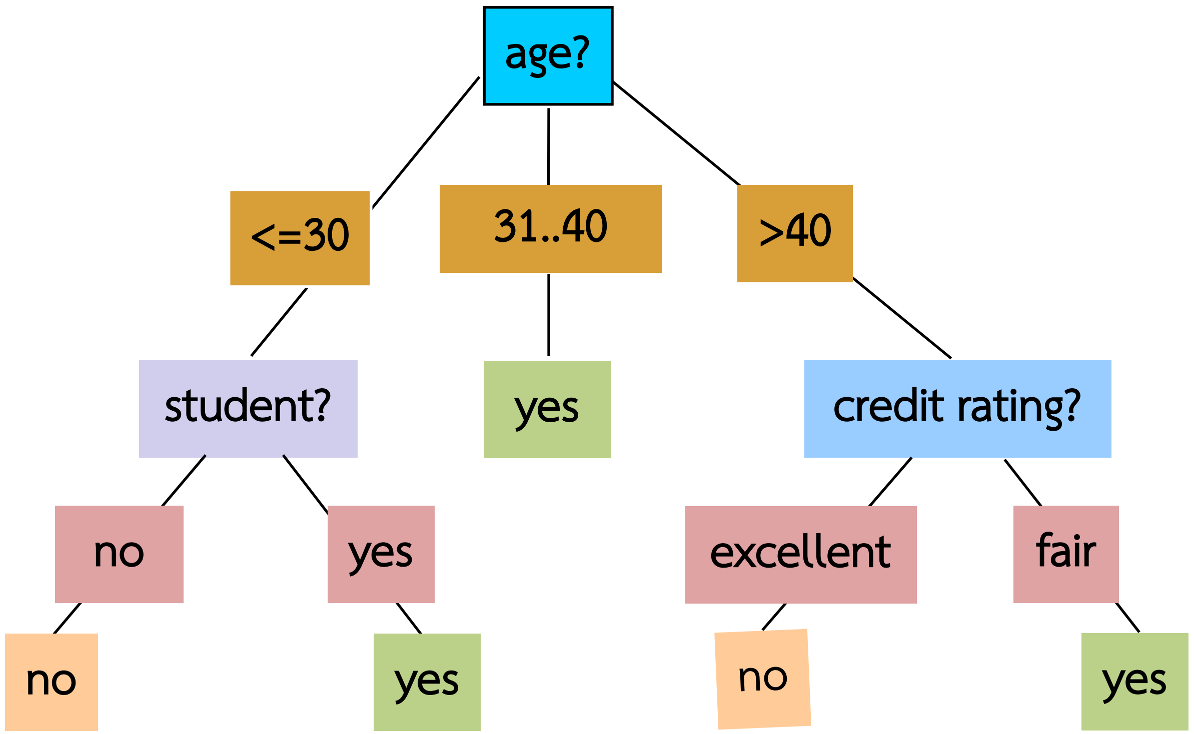
Example

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

จากข้อมูลในตาราง ประกอบด้วย 5 แอตทริบิวต์ คือ

- **age** แสดงอายุ ประกอบด้วย 3 ค่า คือ <=30, 31..40, >40
- **income** แสดงรายได้ประกอบด้วย 3 ค่า คือ high, medium, low
- **student** แสดงสถานะว่าเป็นนักเรียนหรือไม่ ประกอบด้วย 2 ค่า คือ yes, no
- **credit_rating** แสดงอันดับเครดิต ประกอบด้วย 2 ค่า คือ fair, excellent
- **buys_computer** แสดงการซื้อคอมพิวเตอร์ ซึ่งเป็นคลาส ประกอบด้วย 2 ค่า คือ yes, no

Classification – Decision Tree



Example: Rule extraction from our *buys_computer* decision-tree

IF *age* = young AND *student* = *no* THEN *buys_computer* = *no*
IF *age* = young AND *student* = *yes* THEN *buys_computer* = *yes*
IF *age* = mid-age THEN *buys_computer* = *yes*
IF *age* = old AND *credit_rating* = *excellent* THEN *buys_computer* = *no*
IF *age* = old AND *credit_rating* = *fair* THEN *buys_computer* = *yes*

Example 1

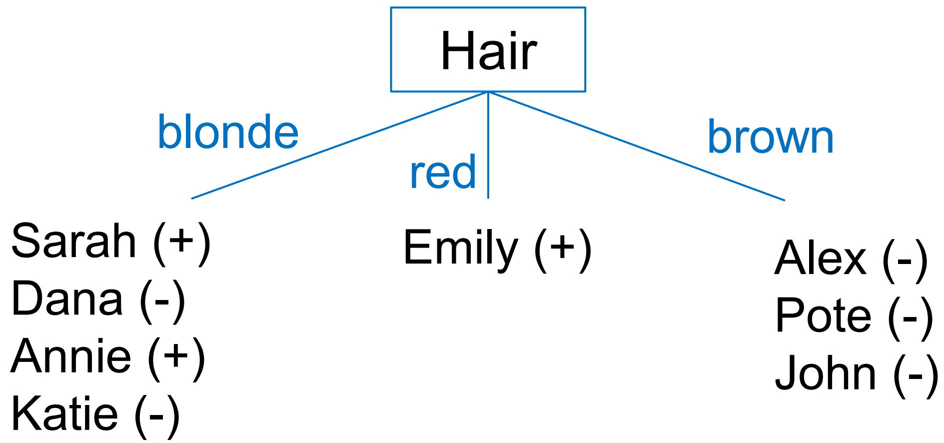
ข้อมูลการศึกษาปัจจัยที่ส่งผลต่อการอาบแดดบริเวณชายหาดที่พบว่าบางคนอาบแดดแล้วสีผิวจะเปลี่ยนเป็นสีแทน (tan) ในขณะที่บางคนได้รับความทรมานจากผิวไหม้

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

การสร้างโมเดล decision tree จะทำการคัดเลือก attribute ที่มีความสัมพันธ์กับ class มากที่สุดขึ้นมาเป็น root node หลังจากนั้นก็จะหา attribute ถัดไปเรื่อยๆ

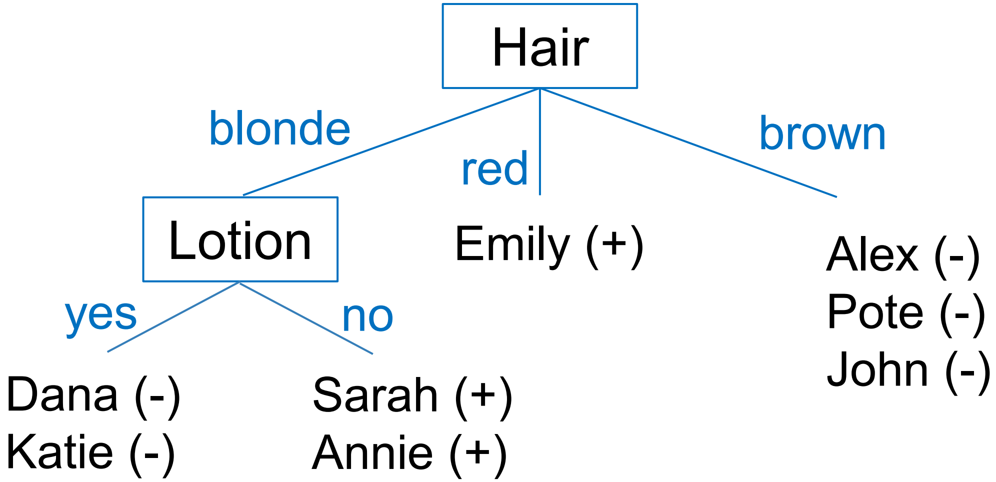
ทดลองเริ่มจาก attribute “Hair”

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none



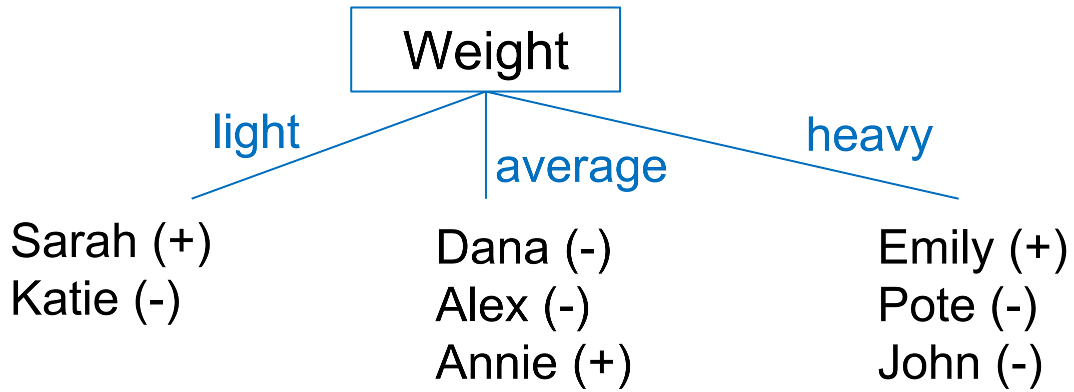
ทดลองเริ่มจาก attribute “Hair”

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none



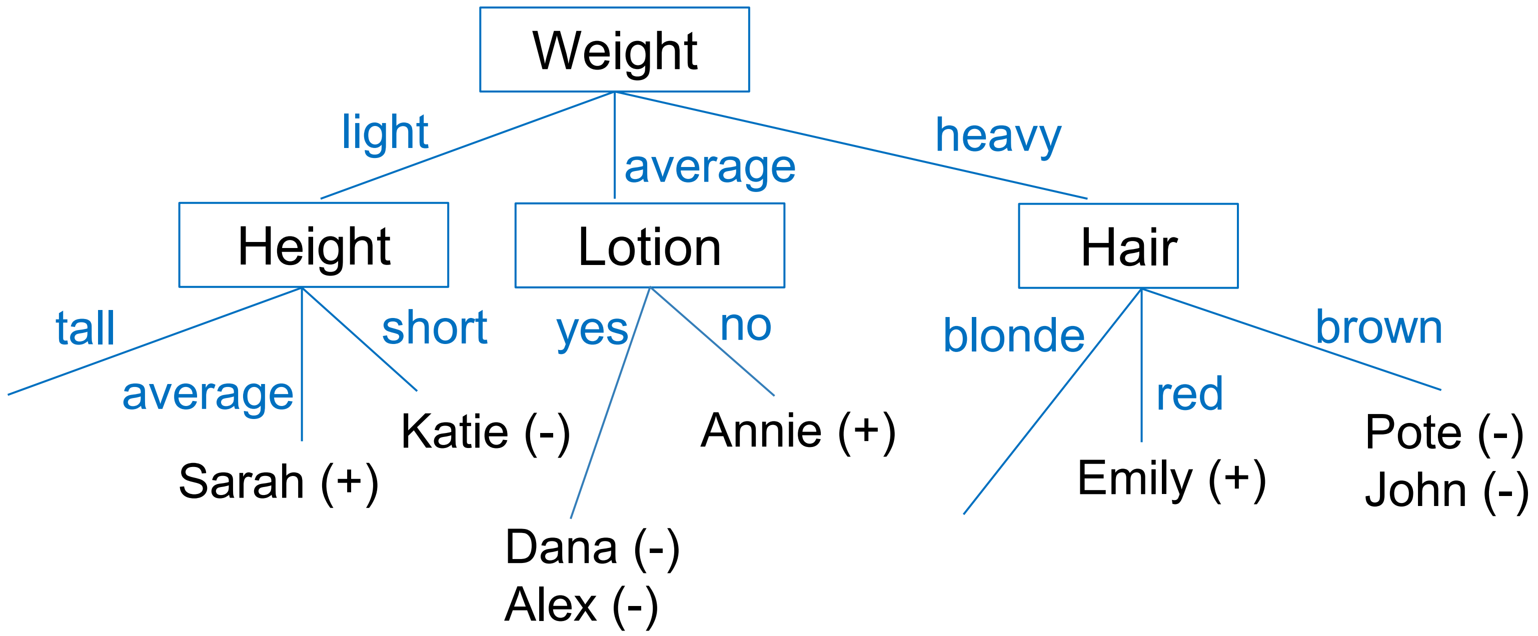
ทดลองเริ่มจาก attribute “Weight”

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none



ทดลองเริ่มจาก attribute "Weight"

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

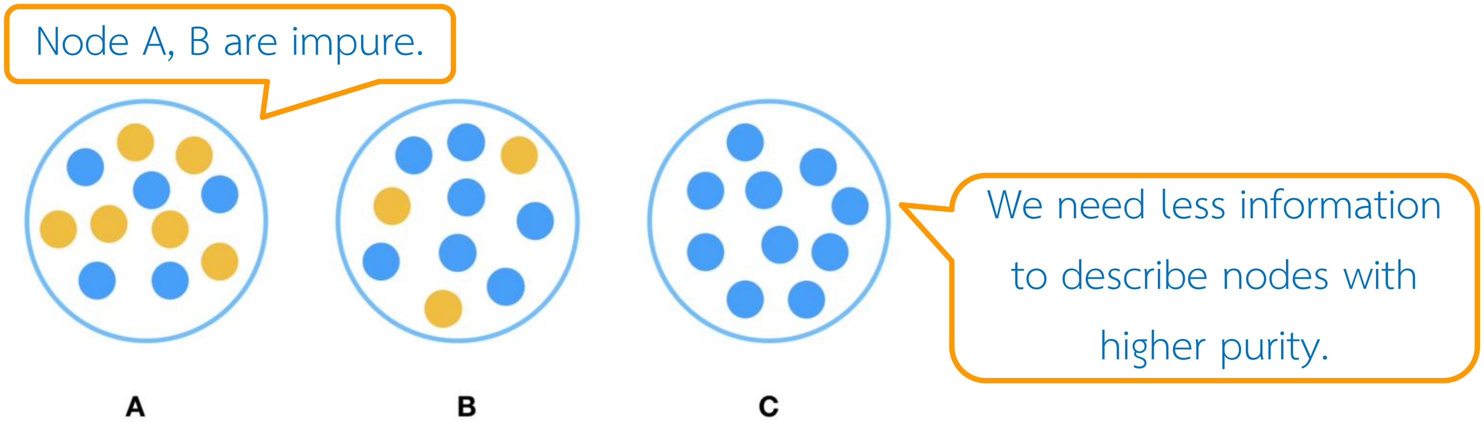


Informative attributes

- Informative attribute provides information that reduces uncertainty about something.
- The better information. The more uncertainty is reduced.
- For example
  - Two friends said that the class is canceled
  - Teacher said that the class is canceled
- Which attribute should be used to segment the dataset?

Purity

- Technically, we would like the group to be as pure as possible.
- Pure means homogeneous with respect to the target variable.
- If some member in the group has a different target, then the group is impure.
- Comparing: which one is the easiest to describe?



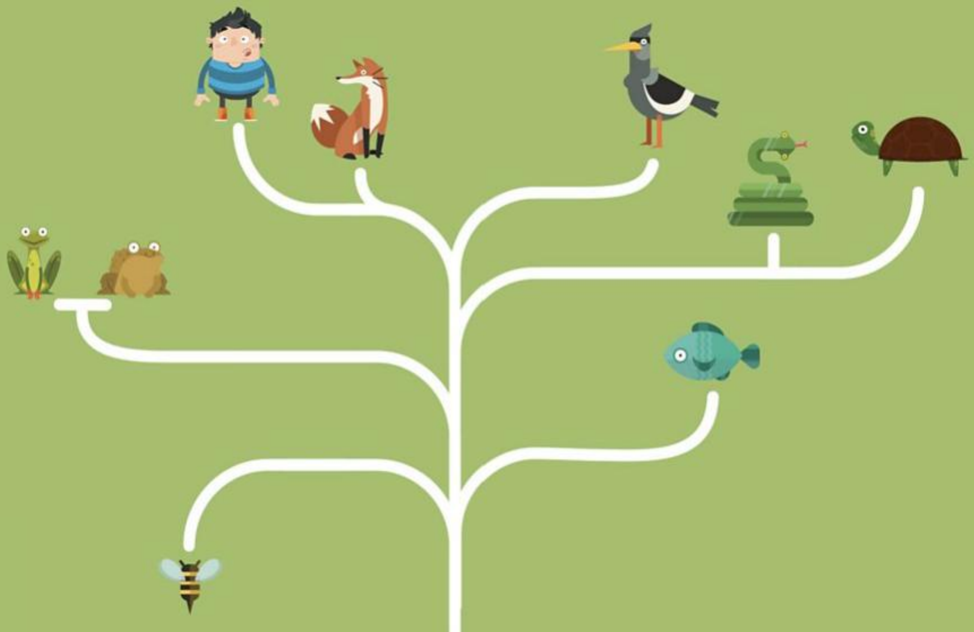
Commonly Used Decision Tree Algorithms

1. ID3 (Iterative Dichotomiser 3)

   - **Main Idea:** The ID3 algorithm uses entropy and information gain to construct decision trees. It iteratively selects the best attribute to split the dataset based on the highest information gain until all data points belong to the same class or no attributes are left.
   - **Limitations:** It cannot handle continuous-valued attributes and may create overfitting trees due to its greedy nature.

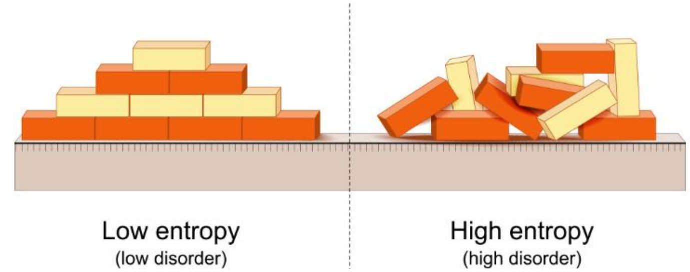
2. C4.5 (Classification and Regression Tree)
3. CART (Classification and Regression Trees)
4. Random Forest
5. Gradient Boosted Trees

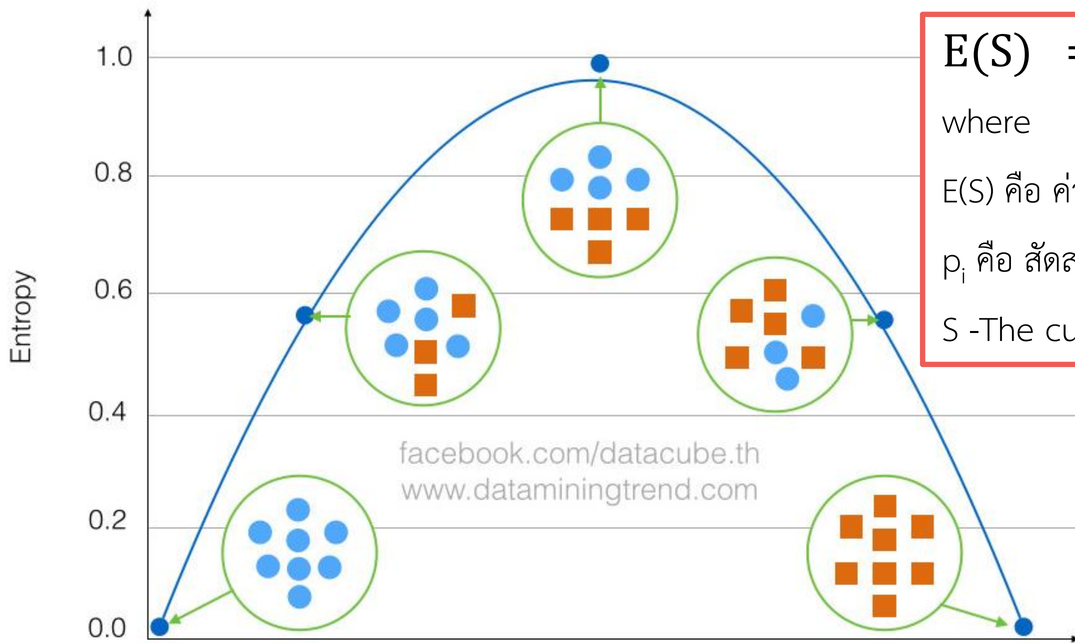
ID3
(Iterative Dichotomiser 3)



Entropy เป็นการวัดความแตกต่างหรือความไม่เป็นระเบียบของข้อมูล

- คุณลักษณะของ Entropy
  - ถ้ามีความ**เหมือนกัน**มากค่า Entropy จะ**ต่ำ**
  - ถ้ามีความ**ต่างกันมาก**ค่า Entropy จะ**สูง**





$$E(S) = \sum -p_i * \log_2(p_i)$$

where

E(S) คือ ค่าเอนโทรปีของชุดข้อมูล S

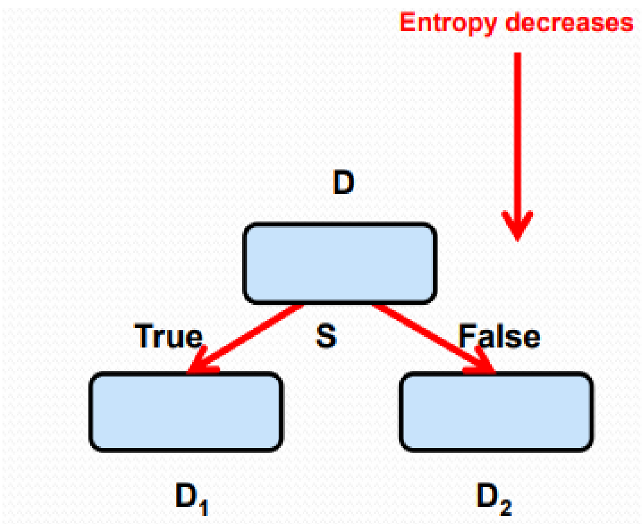
p_i คือ สัดส่วนของข้อมูลในคลาส i กับข้อมูลทั้งหมดในชุดข้อมูล S

S -The current dataset for which entropy is being calculated

ถ้าข้อมูลมีคำตอบหรือคลาสเดียวกัน เช่น เป็นคลาสสีฟ้า หรือ สีส้มทั้งหมดจะมีค่า Entropy ที่ต่ำที่สุด คือ Entropy เท่ากับ 0 แต่ถ้ามีความแตกต่างกันมาก เช่น เป็นคลาสสีฟ้าครึ่งหนึ่งและคลาสสีส้มอีกครึ่งหนึ่งจะมีค่า Entropy สูงสุด คือ Entropy เท่ากับ 1

Information Gain Entropy

- Information gain entropy is a criterion for choosing the features in splitting.
- Information gain is a measure of decrease in entropy after the data is split



Entropy ก่อนแบ่ง

Entropy หลังแบ่ง

$$IG(S, A) = E(S) - \sum [\, p(S|A) * E(S|A) \,]$$

เมื่อ IG(S, A) คือ ค่า Information Gain ของชุดข้อมูล S; attribute A

E(S|A) คือ ค่า Entropy ของชุดข้อมูล S; attribute A

$$E(S) = \sum -p_i * \log_2(p_i)$$

เมื่อ E(S) คือ ค่าเอนโทรปีของชุดข้อมูล S

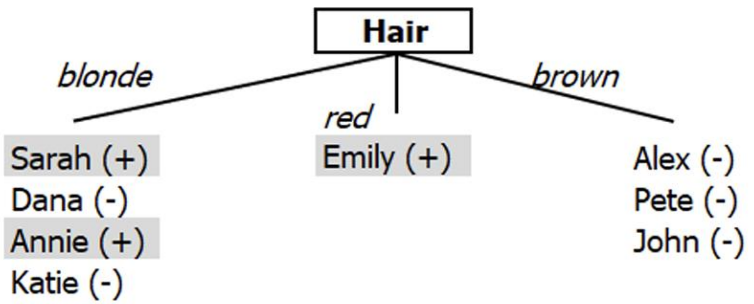
p_i คือ สัดส่วนของข้อมูลในคลาส i กับข้อมูลทั้งหมดในชุดข้อมูล S

We would like to measure how informative an attribute is with respect to target:

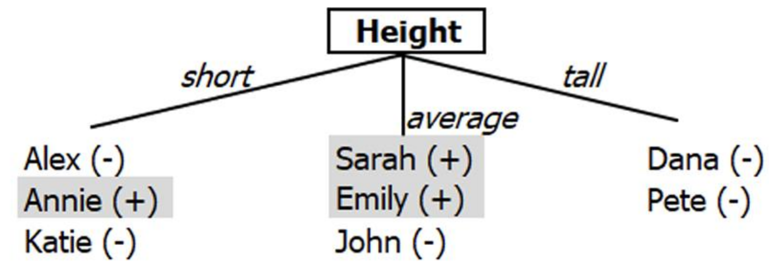
how much gain in information it gives us about the target?

Example 1

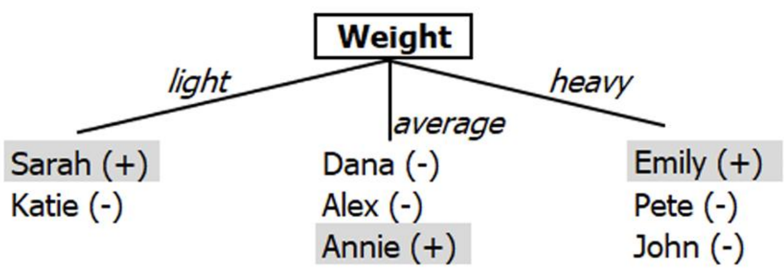
Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none



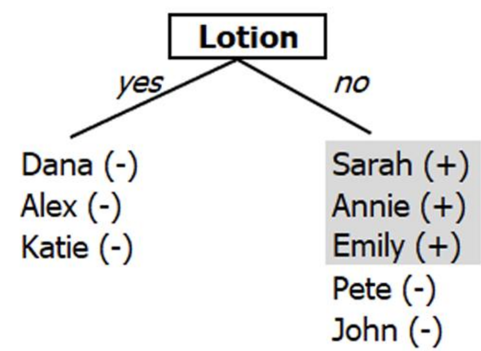
(A)



(B)



(C)



(D)

Example 1

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

คำนวณค่า Entropy เริ่มต้น

$$E(S) = \sum -p_i * \log_2(p_i)$$

$$= [-p(c1)\log_2 p(c1)] + [-p(c2)\log_2 p(c2)]$$

Entropy(c1) Entropy(c2)

$$= [-p(\text{sunburned})\log_2 p(\text{sunburned})] + [-p(\text{none})\log_2 p(\text{none})]$$

$$= [-(3/8)\log_2(3/8)] + [-(5/8)\log_2(5/8)] = 0.954$$

คำนวณค่า Entropy ของ attribute “Hair”

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

$$\mathrm{E}\ (\textbf{Hair}\text{=blonde}) = [-p(\text{sunburned})\log_2 p(\text{sunburned})] + [-p(\text{none})\log_2 p(\text{none})]$$

$$= -[(2/4)\log_2(2/4) + (2/4)\log_2(2/4)] = 1$$

$$\mathrm{E}\ (\textbf{Hair}\text{=brown}) = [-p(\text{sunburned})\log_2 p(\text{sunburned})] + [-p(\text{none})\log_2 p(\text{none})]$$

$$= -[(0/3)\log_2(0/3) + (3/3)\log_2(3/3)] = 0$$

$$\mathrm{E}\ (\textbf{Hair}\text{=red}) = [-p(\text{sunburned})\log_2 p(\text{sunburned})] + [-p(\text{none})\log_2 p(\text{none})]$$

$$= -[(1/1)\log_2(1/1) + (0/1)\log_2(0/1)] = 0$$

คำนวณค่า IG ของ attribute “Hair”

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

$$IG(S, A) = E(S) - \sum [\, p(S|A) * E(S|A) \,]$$

เมื่อ IG(S, A) คือ ค่า Information Gain ของชุดข้อมูล S; attribute A

E(S|A) คือ ค่า Entropy ของชุดข้อมูล S; attribute A

$$\begin{aligned}
&IG(\textbf{Hair}) \\
&= E(S) - [\, p(\textbf{Hair} = \text{blonde}) \times E(\textbf{Hair} = \text{blonde}) \\
&\quad + p(\textbf{Hair} = \text{brown}) \times E(\textbf{Hair} = \text{brown}) \\
&\quad + p(\textbf{Hair} = \text{red}) \times E(\textbf{Hair} = \text{red}) \,] \\
&= 0.954 - [(4/8)\times 1 + (3/8)\times 0 + (1/8)\times 0] \\
&= 0.954 - 0.5 \\
&= 0.454
\end{aligned}$$

คำนวณค่า Entropy ของ attribute "Height"

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

$$- [p(\text{sunburned})\log_2 p(\text{sunburned}) + p(\text{none})\log_2 p(\text{none})]$$

$$E\ (\textbf{Height}=\text{tall}) = -[(0/2)\log_2(0/2) + (2/2)\log_2(2/2)] = 0$$

$$E\ (\textbf{Height}=\text{average}) = -[(2/3)\log_2(2/3) + (1/3)\log_2(1/3)] = 0.918$$

$$E\ (\textbf{Height}=\text{short}) = -[(1/3)\log_2(1/3) + (2/3)\log_2(2/3)] = 0.918$$

คำนวณค่า IG ของ attribute “Height”

$$IG(S, A) \;=\; E(S) - \sum [\, p(S|A) * E(S|A) \,]$$

เมื่อ IG(S, A) คือ ค่า Information Gain ของชุดข้อมูล S; attribute A

E(S|A) คือ ค่า Entropy ของชุดข้อมูล S; attribute A

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

$$IG\ (\mathbf{Height})$$

$$= E(S)\ -\ [\, p(\mathbf{Height}=\text{tall}) \times E(\mathbf{Height}=\text{tall})$$

$$+\ p(\mathbf{Height}=\text{average}) \times E(\mathbf{Height}=\text{average})$$

$$+\ p(\mathbf{Height}=\text{short}) \times E(\mathbf{Height}=\text{short}) \,]$$

$$= 0.954 - [(2/8)\times 0 + (3/8)\times 0.918 + (3/8)\times 0.918]$$

$$= 0.954 - 0.689$$

$$= 0.265$$

คำนวณค่า Entropy ของ attribute “Weight”

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

$$- [p(\text{sunburned})\log_2 p(\text{sunburned}) + p(\text{none})\log_2 p(\text{none})]$$

$$E\ (\textbf{Weight}=\text{light}) = -[(1/2)\log_2(1/2) + (1/2)\log_2(1/2)] = 1$$

$$E\ (\textbf{Weight}=\text{average}) = -[(1/3)\log_2(1/3) + (2/3)\log_2(2/3)] = 0.918$$

$$E\ (\textbf{Weight}=\text{heavy}) = -[(1/3)\log_2(1/3) + (2/3)\log_2(2/3)] = 0.918$$

คำนวณค่า IG ของ attribute “Weight”

$$IG(S, A) \;=\; E(S) - \sum [\, p(S|A) * E(S|A) \,]$$

เมื่อ IG(S, A) คือ ค่า Information Gain ของชุดข้อมูล S; attribute A

E(S|A) คือ ค่า Entropy ของชุดข้อมูล S; attribute A

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

$IG\ (\mathbf{Weight})$

$= E(S)\ -\ [\, p(\mathbf{Weight} = \text{light}) \times E(\mathbf{Weight} = \text{light})$

$\quad + p(\mathbf{Weight} = \text{average}) \times E(\mathbf{Weight} = \text{average})$

$\quad + p(\mathbf{Weight} = \text{heavy}) \times E(\mathbf{Weight} = \text{heavy})\ ]$

$= 0.954 - [(2/8)\times 1 + (3/8)\times 0.918 + (3/8)\times 0.918]$

$= 0.954 - 0.939$

$= 0.015$

คำนวณค่า Entropy ของ attribute “Lotion”

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

$$- [p(\text{sunburned})\log_2 p(\text{sunburned}) + p(\text{none})\log_2 p(\text{none})]$$

$$E\ (\textbf{Lotion}=\text{no}) = -[(3/5)\log_2(3/5) + (2/5)\log_2(2/5)] = 0.971$$

$$E\ (\textbf{Lotion}=\text{yes}) = -[(0/3)\log_2(0/3) + (3/3)\log_2(3/3)] = 0$$

คำนวณค่า IG ของ attribute “Lotion”

$$IG(S, A) = E(S) - \sum [\, p(S|A) * E(S|A) \,]$$

เมื่อ IG(S, A) คือ ค่า Information Gain ของชุดข้อมูล S; attribute A

E(S|A) คือ ค่า Entropy ของชุดข้อมูล S; attribute A

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

IG (**Lotion**)

$= E(S) - [\, p(\textbf{Lotion} = no) \times E(\textbf{Lotion} = no) + p(\textbf{Lotion} = yes) \times E(\textbf{Lotion} = yes) \,]$

$= 0.954 - [(5/8) \times 0.971 + (3/8) \times 0]$

$= 0.954 - 0.607$

$= 0.347$

Information Gain for Each Factor

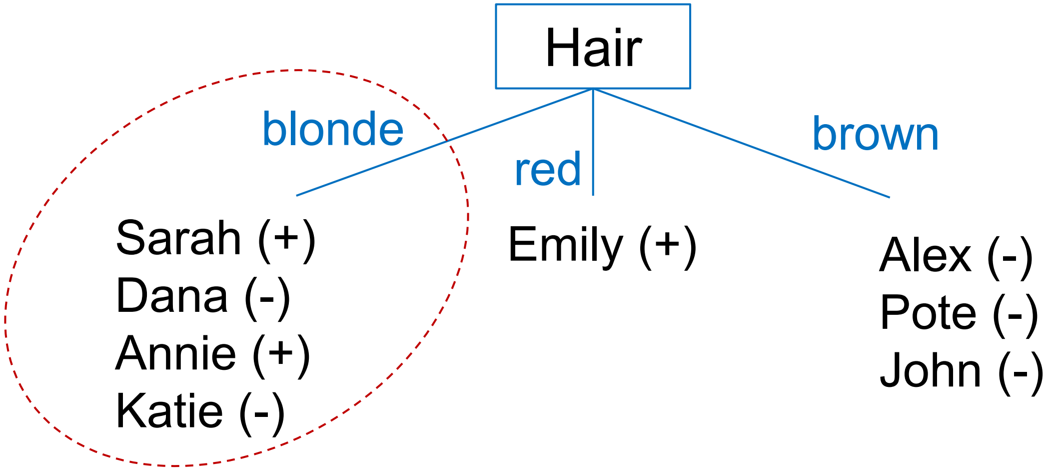
1- IG(Result, Hair) = ...0.454..........

2- IG(Result, Height) = ...0.265............

3- IG(Result, Weight) = ...0.015..........

3- IG(Result, Lotion) = ...0.347..........

Hair factor on decision produces the highest score. That's why, hair decision will appear in the root node of the tree



การแตกกิ่งออกจาก node ที่มี attribute Hair = blonde

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

คำนวณค่า Entropy เริ่มต้น

$E(S) = -p(\textbf{Result}=\text{sunburned}) \times \log2\ p(\textbf{Result}=\text{sunburned}) +$

$-p(\textbf{Result}=\text{none}) \times \log2\ p(\textbf{Result}=\text{none})$

$= -[(2/4)\times\log2(2/4) + (2/4)\times\log2(2/4)]$

$= 1$

คำนวณค่า Entropy ของ attribute Height (เฉพาะที่ Hair=blonde)

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

$$- [p(\text{sunburned})\log_2 p(\text{sunburned}) + p(\text{none})\log_2 p(\text{none})]$$

$$E\ (\textbf{Hair} = \text{blonde}\ \&\ \textbf{Height} = \text{tall}) = -[(0/1)\log_2(0/1) + (1/1)\log_2(1/1)] = 0$$

$$E\ (\textbf{Hair} = \text{blonde}\ \&\ \textbf{Height} = \text{average}) = -[(1/1)\log_2(1/1) + (0/1)\log_2(0/1)] = 0$$

$$E\ (\textbf{Hair} = \text{blonde}\ \&\ \textbf{Height} = \text{short}) = -[(1/2)\log_2(1/2) + (1/2)\log_2(1/2)] = 1$$

คำนวณค่า IG ของ attribute Height (เฉพาะที่ Hair=blonde)

IG(Hair=blonde|Height)

$$IG(S, A) = E(S) - \sum [\, p(S|A) * E(S|A) \,]$$

IG (Hair=blonde|**Height**) = E(S)

– [p(**Hair**=blonde & **Height**=tall) × Entropy(**Hair** = blonde & **Height**=tall)

+p(**Hair**=blonde & **Height**=average) × Entropy(**Hair**=blonde & **Height**=average)

+p(**Hair**=blonde & **Height**=short) × Entropy(**Hair**=blonde & **Height**=short)]

$= 1 - [1/4 \times 0 + 1/4 \times 0 + 2/4 \times 1]$

$= 1 - 0.5$

$= 0.5$

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

คำนวณค่า Entropy ของ attribute Weight (เฉพาะที่ Hair=blonde)

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

$$- [p(\text{sunburned})\log_2 p(\text{sunburned}) + p(\text{none})\log_2 p(\text{none})]$$

$$E\ (\textbf{Hair} = \text{blonde} \ \&\ \textbf{Weight} = \text{light}) = -[(1/2)\log_2(1/2) + (1/2)\log_2(1/2)] = 1$$

$$E\ (\textbf{Hair} = \text{blonde} \ \&\ \textbf{Weight} = \text{average}) = -[(1/2)\log_2(1/2) + (1/2)\log_2(1/2)] = 1$$

$$E\ (\textbf{Hair} = \text{blonde} \ \&\ \textbf{Weight} = \text{heavy}) =$$

คำนวณค่า IG ของ attribute Weight (เฉพาะที่ Hair=blonde)

IG(Hair=blonde|Weight)

$$IG(S, A) = E(S) - \sum [\, p(S|A) * E(S|A) \,]$$

$IG\,(Hair{=}blonde|\mathbf{Weight}) = E(S)$

$- [p(\mathbf{Hair}{=}blonde\ \&\ \mathbf{Weight}{=}light) \times Entropy(\mathbf{Hair} = blonde\ \&\ \mathbf{Weight}{=}light)$

$+p(\mathbf{Hair}{=}blonde\ \&\ \mathbf{Weight}{=}average) \times Entropy(\mathbf{Hair}{=}blonde\ \&\ \mathbf{Weight}{=}average)]$

$= 1 - [2/4{\times}1 + 2/4{\times}1]$

$= 1 - 1$

$= 0$

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

คำนวณค่า Entropy ของ attribute Lotion (เฉพาะที่ Hair=blonde)

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

$$- [p(\text{sunburned})\log_2 p(\text{sunburned}) + p(\text{none})\log_2 p(\text{none})]$$

$$E\,(\textbf{Hair} = \text{blonde}\ \&\ \textbf{Lotion} = \text{no}) \quad = \quad -[2/2)\log_2(2/2) + (0/2)\log_2(0/2)] \ = \ 0$$

$$E\,(\textbf{Hair} = \text{blonde}\ \&\ \textbf{Lotion} = \text{yes}) \quad = \quad -[(0/2)\log_2(0/2) + (2/2)\log_2(2/2)] \ = 0$$

คำนวณค่า IG ของ attribute Lotion (เฉพาะที่ Hair=blonde)

IG(Hair=blonde|Lotion)

$$IG(S, A) = E(S) - \sum [\, p(S|A) * E(S|A) \,]$$

IG (Hair=blonde|**Lotion**) = E(S)

– [p(**Hair**=blonde & **Height**=tall) × Entropy(**Hair** = blonde & **Height**=tall)

+p(**Hair**=blonde & **Height**=average) × Entropy(**Hair**=blonde & **Height**=average)]

$= 1 - [2/2\times0 + 2/2\times0]$

$= 1 - 0$

$= 1$

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

Blonde hair on decision

There are 4 instances for blonde hair. Decision would be probably 2/4 percent sunburned, 2/4 percent none.

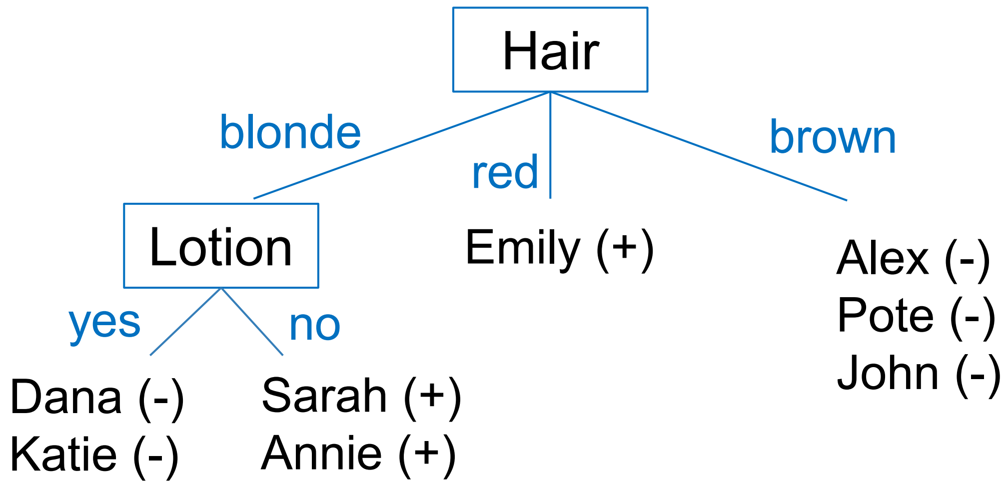
1- IG(Hair=blonde|Height) = 0.5

2- IG(Hair=blonde|Weight) = 0

3- IG(Hair=blonde|Lotion) = 1

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

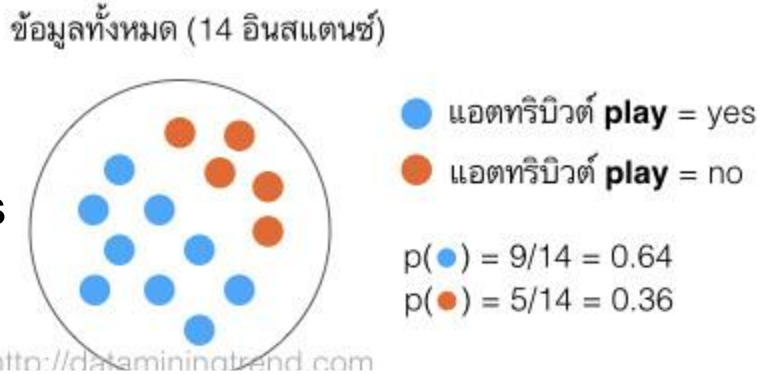
Now, **Lotion** is the decision because it produces the highest score if hair were blonde



Example 2

Day	Outlook	Temperature	Humidity	Windy	Play Tennis Decision
1	sunny	hot	high	false	no
2	sunny	hot	high	true	no
3	overcast	hot	high	false	yes
4	rain	mild	high	false	yes
5	rain	cool	normal	false	yes
6	rain	cool	normal	true	no
7	overcast	cool	normal	true	yes
8	sunny	mild	high	false	no
9	sunny	cool	normal	false	yes
10	rain	mild	normal	false	yes
11	sunny	mild	normal	true	yes
12	overcast	mild	high	true	yes
13	overcast	hot	normal	false	yes
14	rain	mild	high	true	no

0. คำนวณค่า Entropy เริ่มต้น



- Decision column consists of **14 instances** and includes two labels: yes and no.
- There are 9 decisions labeled yes, and 5 decisions labeled no.

$$\text{Entropy}(S) = \sum -p(I) * \log_2 p(I)$$

$$= \underbrace{[-p(c1)\log_2 p(c1)]}_{\text{Entropy}(c1)} + \underbrace{[-p(c2)\log_2 p(c2)]}_{\text{Entropy}(c2)}$$

$$\text{Entropy}(\text{Decision}) = -\,p(\text{Yes}) \,.\, \log_2 p(\text{Yes}) - p(\text{No}) \,.\, \log_2 p(\text{No})$$

$$\text{Entropy}(\text{Decision}) = -\,(9/14) \,.\, \log_2(9/14) - (5/14) \,.\, \log_2(5/14) = 0.940$$

https://sefiks.com/2017/11/20/a-step-by-step-id3-decision-tree-example/

1.1.1 คำนวณค่า Entropy ของ attribute Outlook

$$\text{Entropy} = [-p(c1)\log_2 p(c1)] + [-p(c2)\log_2 p(c2)]$$

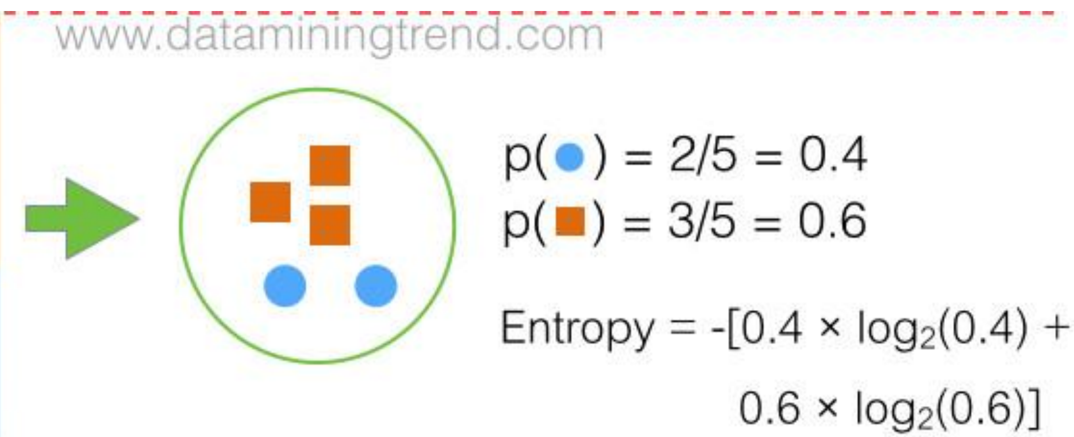
$$\text{Entropy}\ (\textbf{Outlook}=\text{sunny}) = -p(\text{yes}) \times \log_2 p(\text{yes}) - p(\text{no}) \times \log_2 p(\text{no})$$

$$= -[0.4 \times \log_2(0.4) + 0.6 \times \log_2(0.6)]$$

$$= -[(0.4)(-1.32) + (0.6)(-0.74)]$$

$$= 0.97$$

1	sunny	no
2	sunny	no
8	sunny	no
9	sunny	yes
11	sunny	yes



https://datacubeth.ai/decision-tree/

1.1.1 คำนวณค่า Entropy ของ attribute Outlook

$$\text{Entropy} = [-p(c1)\log_2 p(c1)] + [-p(c2)\log_2 p(c2)]$$

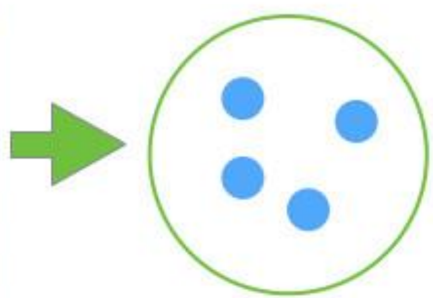
$$\text{Entropy}\ (\textbf{Outlook}=\text{overcast}) = -p(\text{yes}) \times \log_2 p(\text{yes}) - p(\text{no}) \times \log_2 p(\text{no})$$

$$= -[1 \times \log_2(1) + 0 \times \log_2(0)]$$

$$= -[(1)(0) + (0)(\log_2(0))]$$

$$= 0$$

ID	Outlook	Play
3	overcast	yes
7	overcast	yes
12	overcast	yes
13	overcast	yes



p(●) = 4/4 = 1.0

p(■) = 0/4 = 0.0

Entropy = -[1.0 × $\log_2(1.0)$ + 0.0 × $\log_2(0.0)$]

1.1.1 คำนวณค่า Entropy ของ attribute Outlook

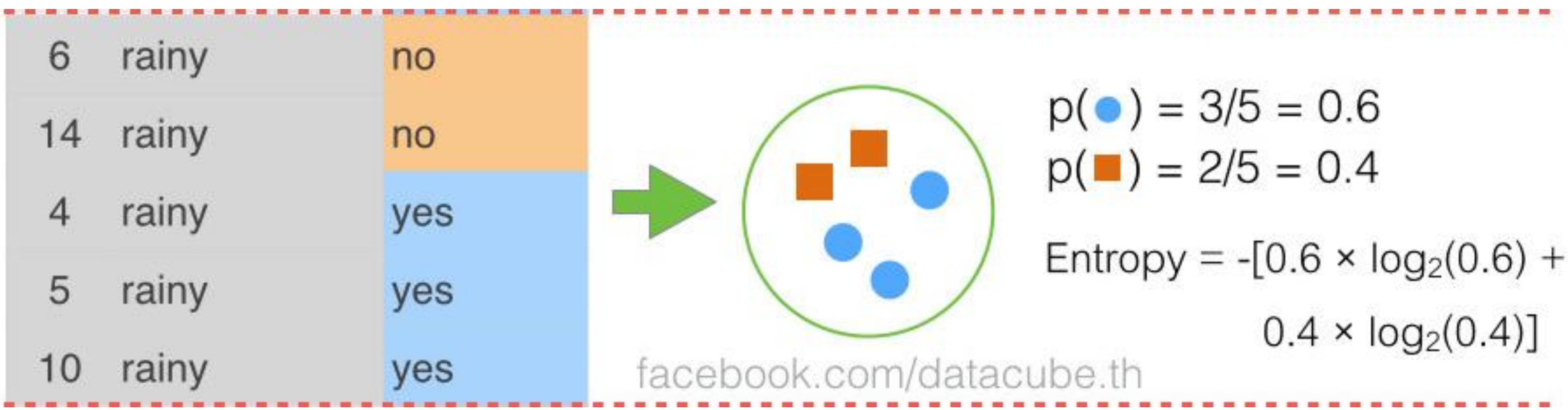
$$\text{Entropy} = [-p(c1)\log_2 p(c1)] + [-p(c2)\log_2 p(c2)]$$

$$\text{Entropy }(\textbf{Outlook}{=}\text{rain}) = -p(\text{yes}) \times \log_2 p(\text{yes}) - p(\text{no}) \times \log_2 p(\text{no})$$

$$= -[0.6 \times \log_2(0.6) + 0.4 \times \log_2(0.4)]$$

$$= -[(0.6)(-0.74) + (0.4)(-1.32)]$$

$$= \ 0.97$$



https://datacubeth.ai/decision-tree/

1.1.2 คำนวณค่า IG ของ attribute Outlook

IG(parent, child) = Entropy(parent) – ∑ [p(parent|child) * Entropy(parent|child)]

= Entropy(parent) – [p(c1) x Entropy(c1) + p(c2) x Entropy(c2) + ...]

Information Gain (**Outlook**)

= Entropy(initial) – [P(**Outlook**=sunny) × Entropy(**Outlook**=sunny)

+ P(**Outlook**=overcast) × Entropy(**Outlook**=overcast)

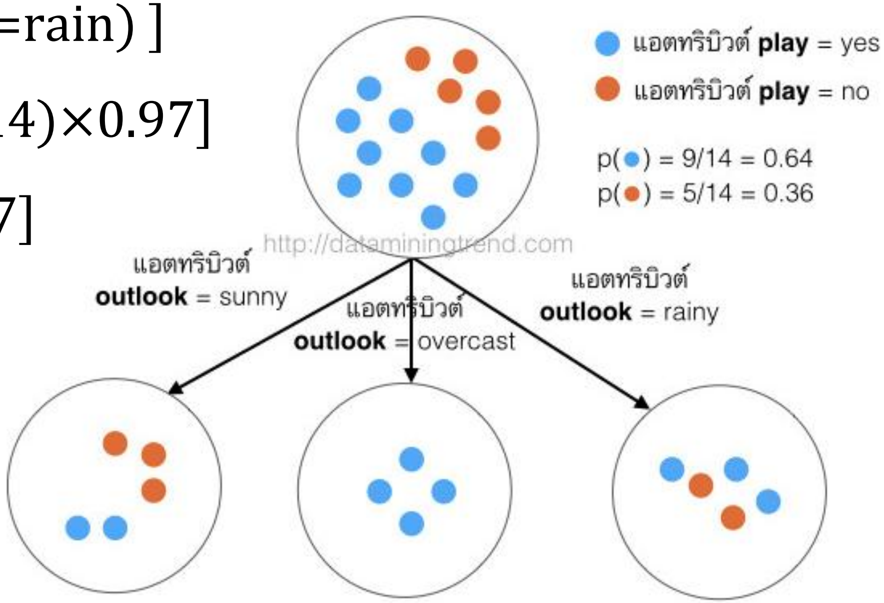
+ P(**Outlook**=rain) × Entropy(**Outlook**=rain)]

= 0.94 – [(5/14)×0.97 + (4/14)×0 + (5/14)×0.97]

= 0.94 – [0.36×0.97 + 0.29×0 + 0.36×0.97]

= 0.94 – 0.7

= 0.24



https://datacubeth.ai/decision-tree/

1.2.1 คำนวณค่า Entropy ของ attribute Temperature

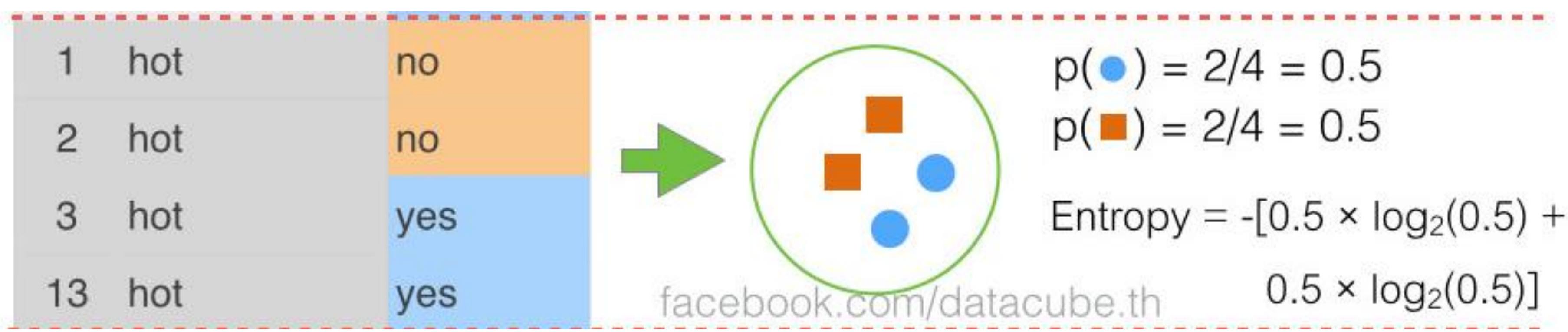
$$\text{Entropy} = [\text{-p(c1)}\log_2\text{p(c1)}] + [\text{-p(c2)}\log_2\text{p(c2)}]$$

$$\text{Entropy } (\textbf{Temp}{=}\text{hot}) = -\text{p(yes)} \times \log_2 \text{p(yes)} - \text{p(no)} \times \log_2 \text{p(no)}$$

$$= -[0.5 \times \log_2(0.5) + 0.5 \times \log_2(0.5)]$$

$$= -[(0.)(\text{-}1) + (0.5)(\text{-}1)]$$

$$= 1.00$$



https://datacubeth.ai/decision-tree/

1.2.1 คำนวณค่า Entropy ของ attribute Temperature

$$\text{Entropy} = [-p(c1)\log_2 p(c1)] + [-p(c2)\log_2 p(c2)]$$

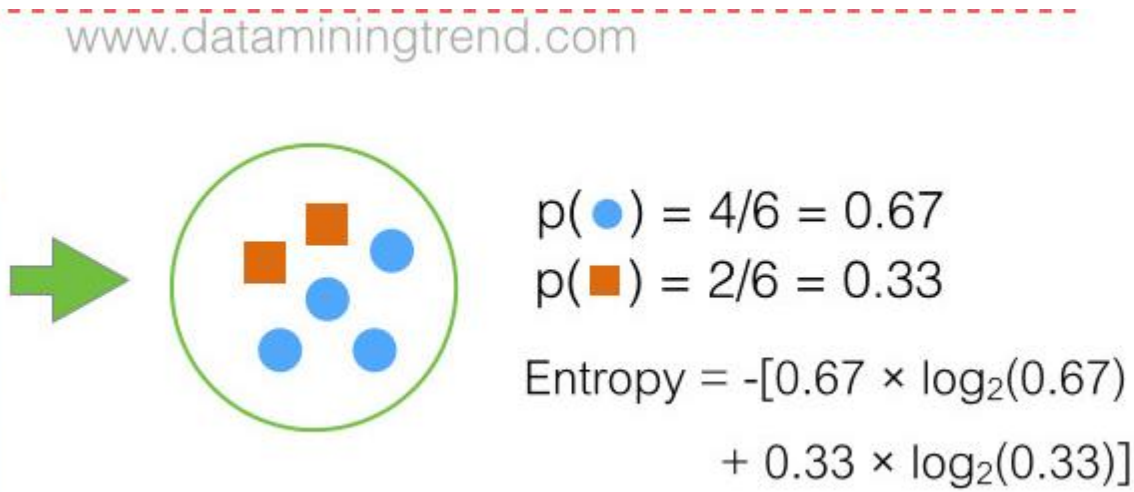
$$\text{Entropy } (\textbf{Temp}=\text{mild}) = -p(\text{yes}) \times \log_2 p(\text{yes}) - p(\text{no}) \times \log_2 p(\text{no})$$

$$= -[0.67 \times \log_2(0.67) + 0.33 \times \log_2(0.33)]$$

$$= -[(0.67)(-0.58) + (0.33)(-1.60)]$$

$$= 0.91$$

8	mild	no
14	mild	no
4	mild	yes
10	mild	yes
11	mild	yes
12	mild	yes



https://datacubeth.ai/decision-tree/

1.2.1 คำนวณค่า Entropy ของ attribute Temperature

$$\text{Entropy} = [-p(c1)\log_2 p(c1)] + [-p(c2)\log_2 p(c2)]$$

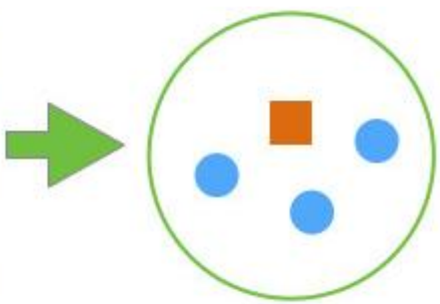
$$\text{Entropy } (\textbf{Temp}{=}\text{cool}) = -p(\text{yes}) \times \log_2 p(\text{yes}) - p(\text{no}) \times \log_2 p(\text{no})$$

$$= -[0.75 \times \log_2(0.75) + 0.25 \times \log_2(0.25)]$$

$$= -[(0.75)(-0.42) + (0.25)(-2)]$$

$$= 0.81$$

6	cool	no
5	cool	yes
7	cool	yes
9	cool	yes



p(●) =3/4 = 0.75

p(■) = 1/4 = 0.25

$$\text{Entropy} = -[0.75 \times \log_2(0.75) + 0.25 \times \log_2(0.25)]$$

https://datacubeth.ai/decision-tree/

1.2.2 คำนวณค่า IG ของ attribute Temperature

IG(parent, child) = Entropy(parent) – ∑ [p(parent|child) * Entropy(parent|child)]

= Entropy(parent) – [p(c1) x Entropy(c1) + p(c2) x Entropy(c2) + ...]

Information Gain (**Temp**)

= Entropy(initial) – [P(**Temp** =hot) × Entropy(**Temp** =hot)

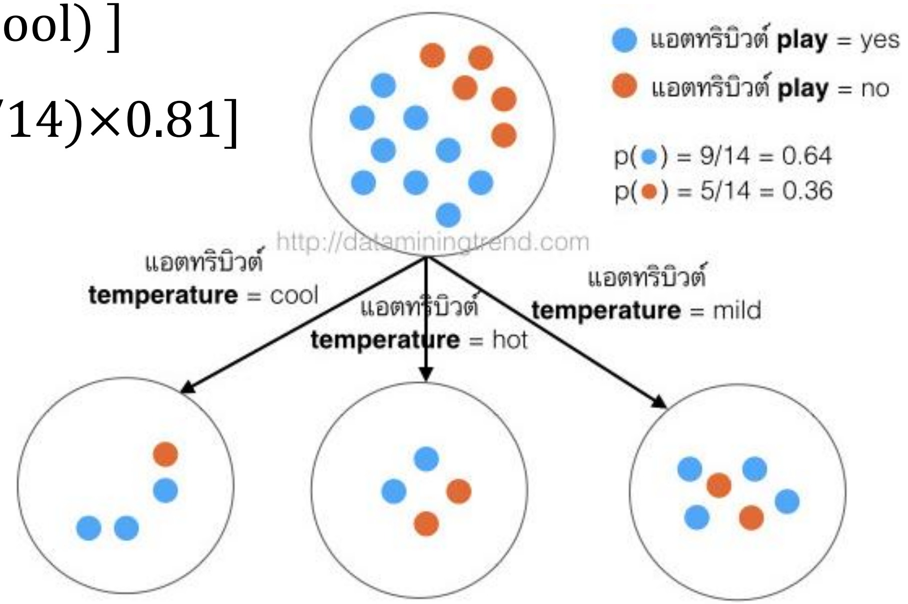
+ P(**Temp** =mild) × Entropy(**Temp** =mild)

+ P(**Temp** =cool) × Entropy(**Temp** =cool)]

= 0.94 – [(4/14)×1 + (6/14)×0.91 + (4/14)×0.81]

= 0.94 – 0.91

= 0.03



https://datacubeth.ai/decision-tree/

Individual Assignment

แสดงวิธีคำนวณค่าที่เหลือ

1- Information Gain(Humidity)

2- Information Gain(Windy)

https://nulpointerexception.com/2017/12/16/a-tutorial-to-understand-decision-tree-id3-learning-algorithm/

1.3.1 คำนวณค่า Entropy ของ attribute Humidity

$$\text{Entropy} = [-p(c1)\log_2 p(c1)] + [-p(c2)\log_2 p(c2)]$$

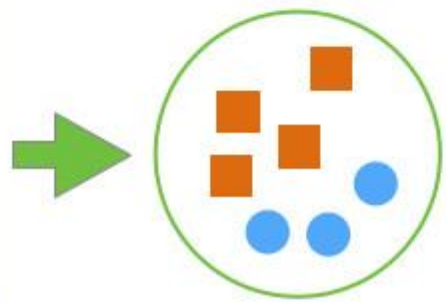
$$\text{Entropy } (\textbf{Humidity} = \text{high}) = -p(\text{yes}) \times \log_2 p(\text{yes}) - p(\text{no}) \times \log_2 p(\text{no})$$

$$= -[0.43 \times \log_2(0.43) + 0.57 \times \log_2(0.57)]$$

$$= -[(0.43)(-1.22) + (0.57)(-0.81)]$$

$$= 0.99$$

1	high	no
2	high	no
8	high	no
14	high	no
3	high	yes
4	high	yes
12	high	yes



p(●) = 3/7 = 0.43

p(■) = 4/7 = 0.57

Entropy = -[0.43 × log2(0.43)

+ 0.57 × log2(0.57)]

facebook.com/datacube.th

1.3.1 คำนวณค่า Entropy ของ attribute Humidity

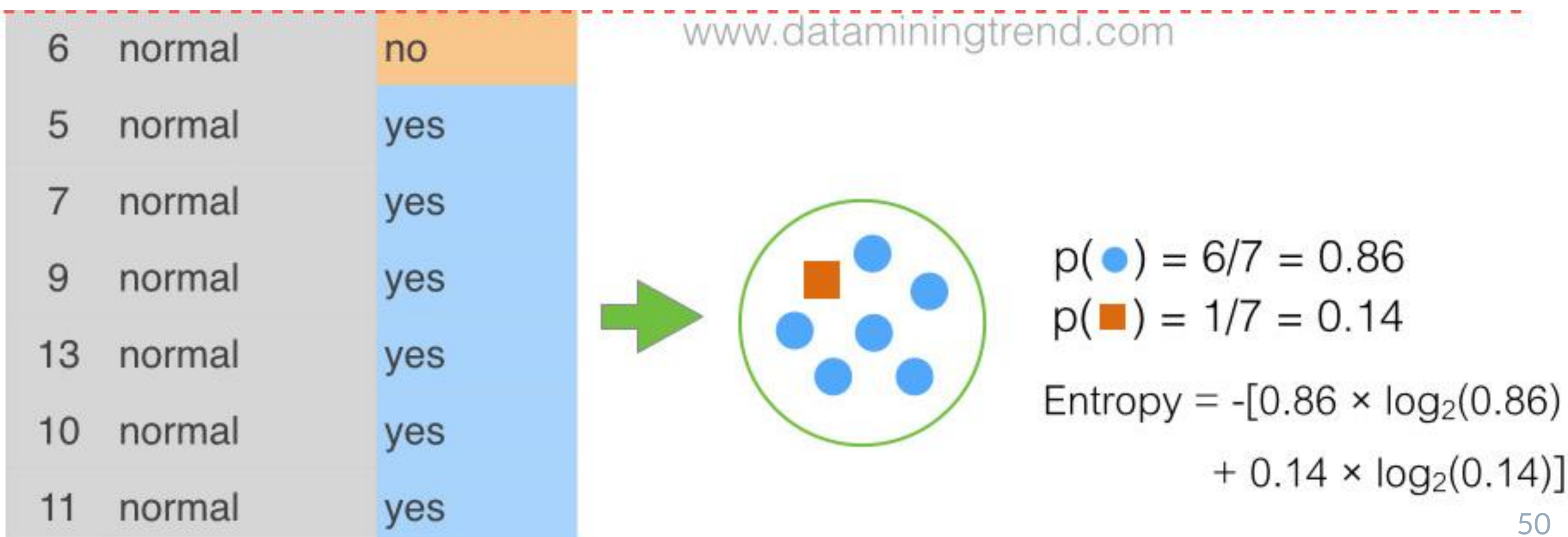
$$\text{Entropy} = -[p(c1)\log_2 p(c1)] + [p(c2)\log_2 p(c2)]$$

$$\text{Entropy } (\textbf{Humidity} = \text{normal}) = -[p(yes) \times \log_2 p(yes) + p(no) \times \log_2 p(no)]$$

$$= -[0.86 \times \log_2(0.86) + 0.14 \times \log_2(0.14)]$$

$$= -[(0.86)(-0.22) + (0.14)(-2.84)]$$

$$= 0.59$$



1.3.2 คำนวณค่า IG ของ attribute Humidity

IG(parent, child) $= \text{Entropy(parent)} - \sum [\, p(\text{parent}|\text{child}) * \text{Entropy(parent}|\text{child}) \,]$

$= \text{Entropy(parent)} - [\, p(c1) \times \text{Entropy}(c1) + p(c2) \times \text{Entropy}(c2) + \ldots \,]$

Information Gain (**Humidity**)

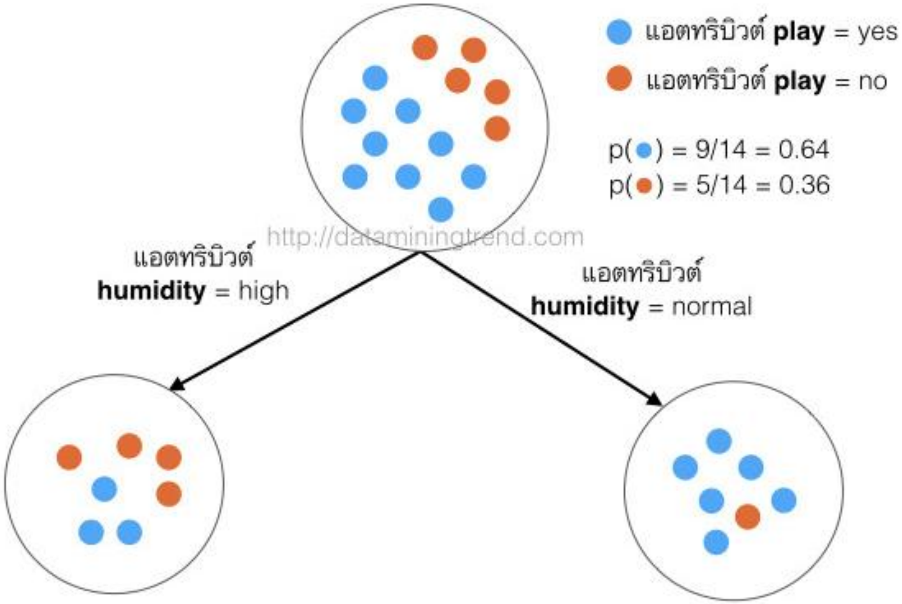
= Entropy(initial) – [P(**Humidity** =high) × Entropy(**Humidity** =high)

+ P(**Humidity** =normal) × Entropy(**Humidity** =normal)]

$= 0.94 - [(7/14)\times 0.99 + (7/14)\times 0.59]$

$= 0.94 - 0.79$

$= 0.15$



https://datacubeth.ai/decision-tree/

1.4.1 คำนวณค่า Entropy ของ attribute Wind

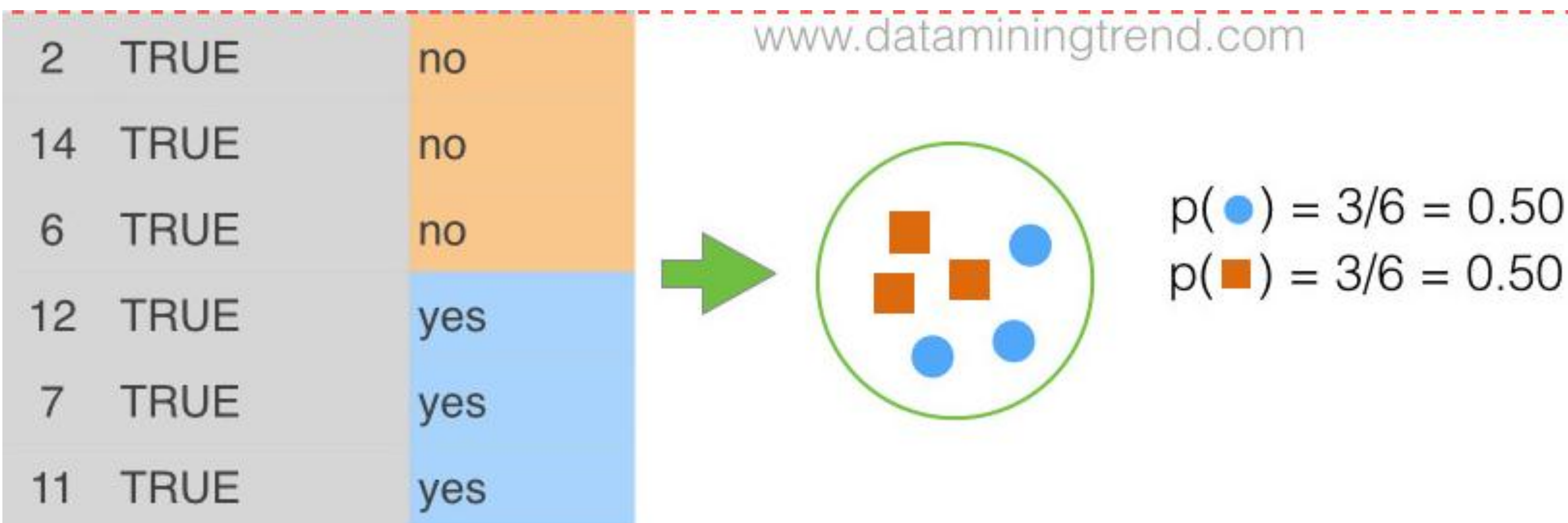
$$\text{Entropy} = [-p(c1)\log_2 p(c1)] + [-p(c2)\log_2 p(c2)]$$

$$\text{Entropy } (\mathbf{Wind} = \text{true}) = -p(\text{yes}) \times \log_2 p(\text{yes}) - p(\text{no}) \times \log_2 p(\text{no})$$

$$= -[0.5 \times \log_2(0.5) + 0.5 \times \log_2(0.5)]$$

$$= -[(0.5)(-0.1) + (0.5)(-0.1)]$$

$$= 1.0$$



1.4.1 คำนวณค่า Entropy ของ attribute Wind

$$\text{Entropy} = [-p(c1)\log_2 p(c1)] + [-p(c2)\log_2 p(c2)]$$

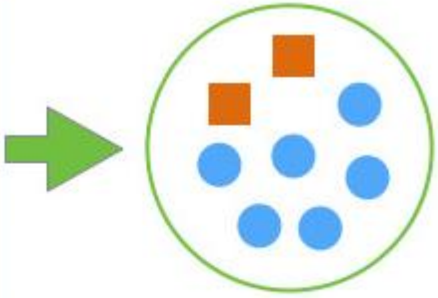
$$\text{Entropy }(\textbf{Wind} = \text{false}) = -p(\text{yes}) \times \log_2 p(\text{yes}) - p(\text{no}) \times \log_2 p(\text{no})$$

$$= -[0.75 \times \log_2(0.75) + 0.25 \times \log_2(0.25)]$$

$$= -[(0.75)(-0.42) + (0.25)(-0.2)]$$

$$= 0.81$$

1	FALSE	no
8	FALSE	no
3	FALSE	yes
4	FALSE	yes
5	FALSE	yes
9	FALSE	yes
13	FALSE	yes
10	FALSE	yes



p(●) = 6/8 = 0.75

p(■) = 2/8 = 0.25

1.4.2 คำนวณค่า IG ของ attribute Wind

IG(parent, child) = Entropy(parent) – ∑ [p(parent|child) * Entropy(parent|child)]

= Entropy(parent) – [p(c1) x Entropy(c1) + p(c2) x Entropy(c2) + …]

Information Gain (**Windy**)

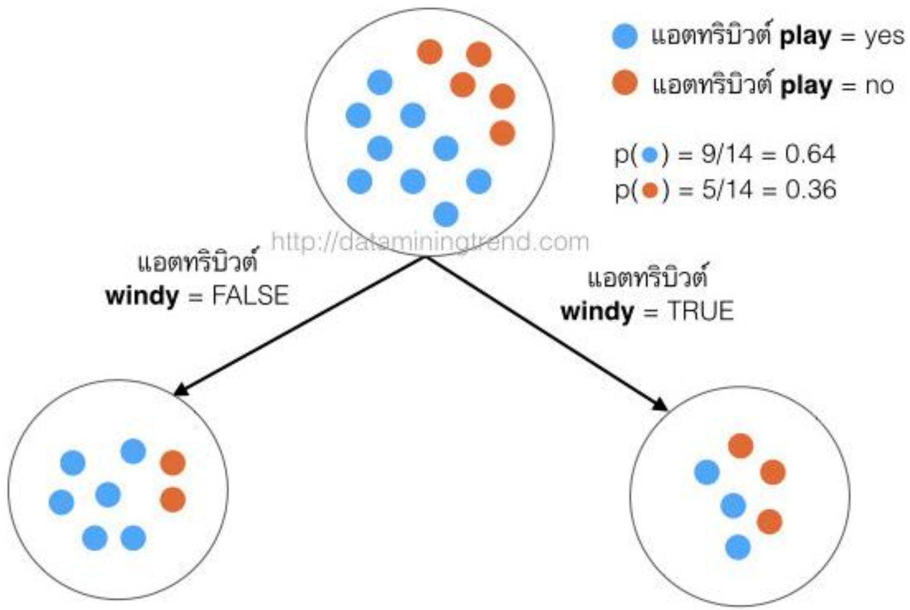
= Entropy(initial) – [P(**Windy**=true) × Entropy(**Windy**=true)

+ P(**Windy**=false) × Entropy(**Windy**=false)]

= 0.94 – [(6/14)×1 + (8/14)×0.58]

= 0.94 – 0.89

= 0.05



https://datacubeth.ai/decision-tree/

Information Gain for Each Factor

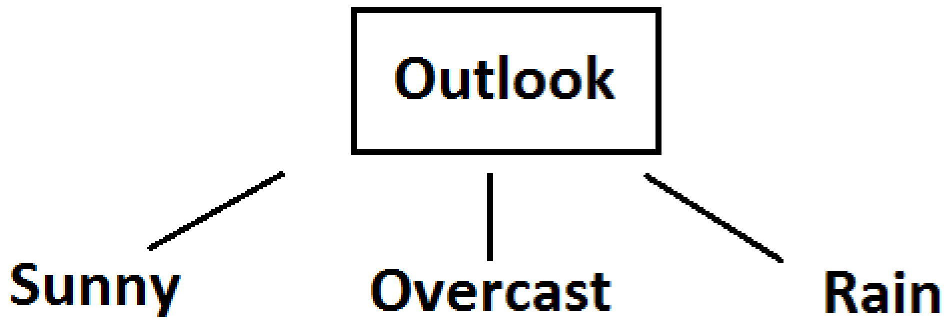
1- Gain(Decision, Outlook) = ...0.24..........

2- Gain(Decision, Temperature) = ...0.03............

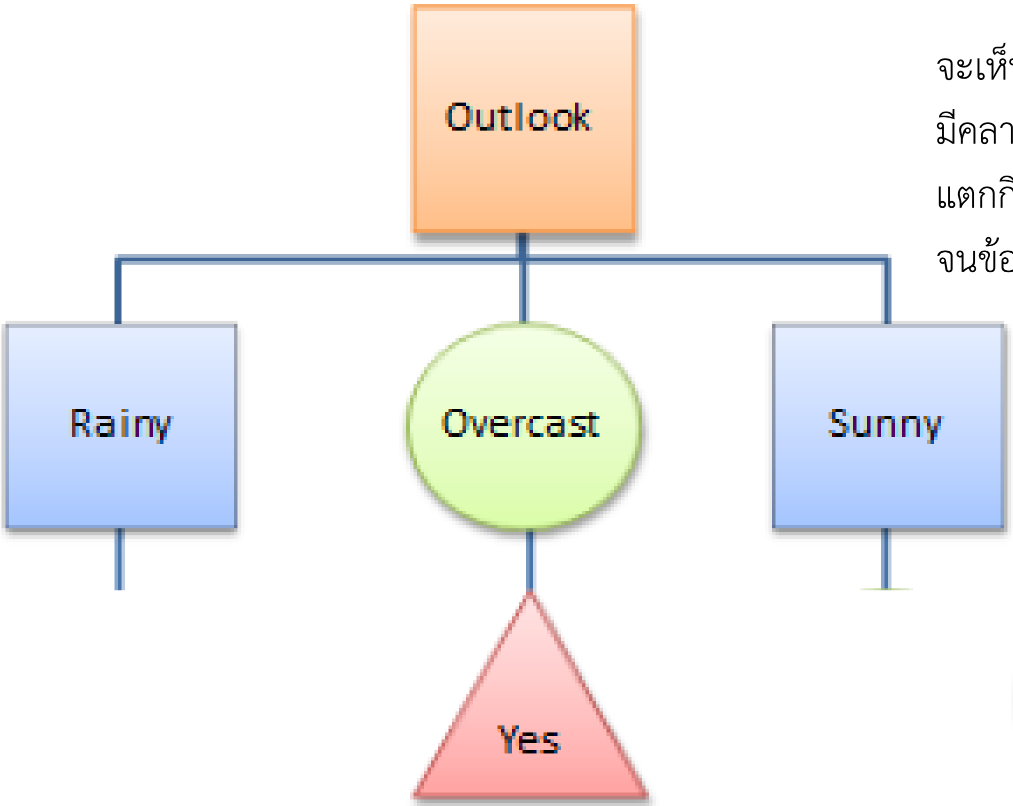
3- Gain(Decision, Humidity) = ...0.15.........

4- Gain(Decision, Windy) = ...0.05.........

Outlook factor on decision produces the highest score. That's why, outlook decision will appear in the root node of the tree



Day	Outlook	Temperature	Humidity	Windy	Play Tennis Decision
3	overcast	hot	high	false	yes
7	overcast	cool	normal	true	yes
12	overcast	mild	high	true	yes
13	overcast	hot	normal	false	yes



จะเห็นว่าข้อมูลที่อยู่ในโหนดที่แอตทริบิวต์ **outlook**=overcast มีคลาสเดียวกันหมดคือ **play**=yes ดังนั้นโหนดนี้ไม่ต้องทำการแตกกิ่งต่อไปอีกแล้ว แต่โหนดอื่นๆ จะต้องทำการแตกกิ่งออกไปจนข้อมูลในแต่ละโหนดมีคลาสคำตอบเดียวกัน

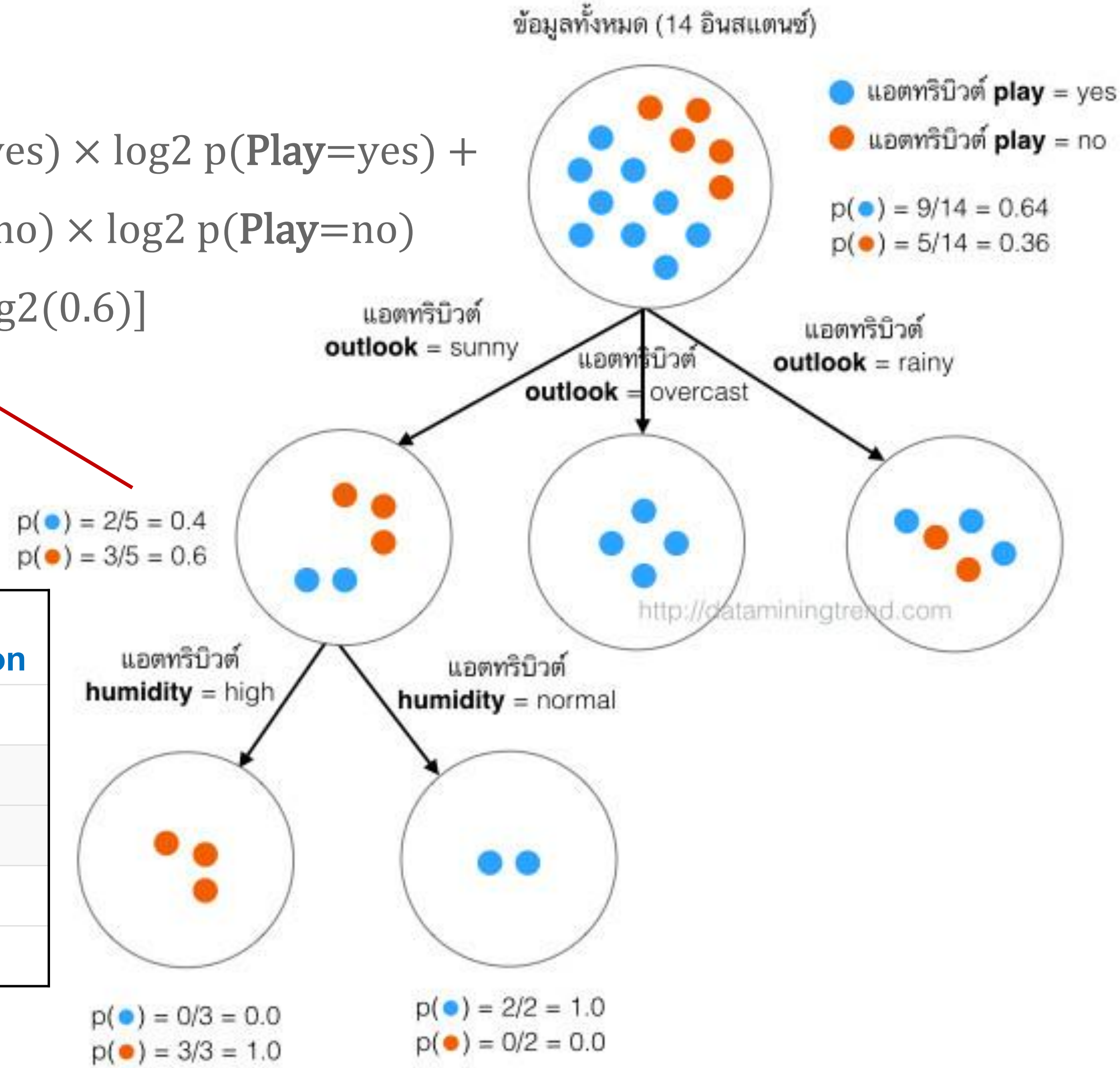
ตัวอย่างการแตกกิ่งออกจากโหนดที่มีแอตทริบิวต์ outlook = sunny

คำนวณค่า Entropy เริ่มต้น

$$\text{entropy (parent)} = -p(\textbf{Play}=\text{yes}) \times \log2\ p(\textbf{Play}=\text{yes}) + $$
$$-p(\textbf{Play}=\text{no}) \times \log2\ p(\textbf{Play}=\text{no})$$

$$= -[0.4 \times \log2(0.4) + 0.6 \times \log2(0.6)]$$

$$= 0.97$$



Day	Outlook	Humidity	Play Decison
1	sunny	high	no
2	sunny	high	no
8	sunny	high	no
9	sunny	normal	yes
11	sunny	normal	yes

https://datacubeth.ai/decision-tree/

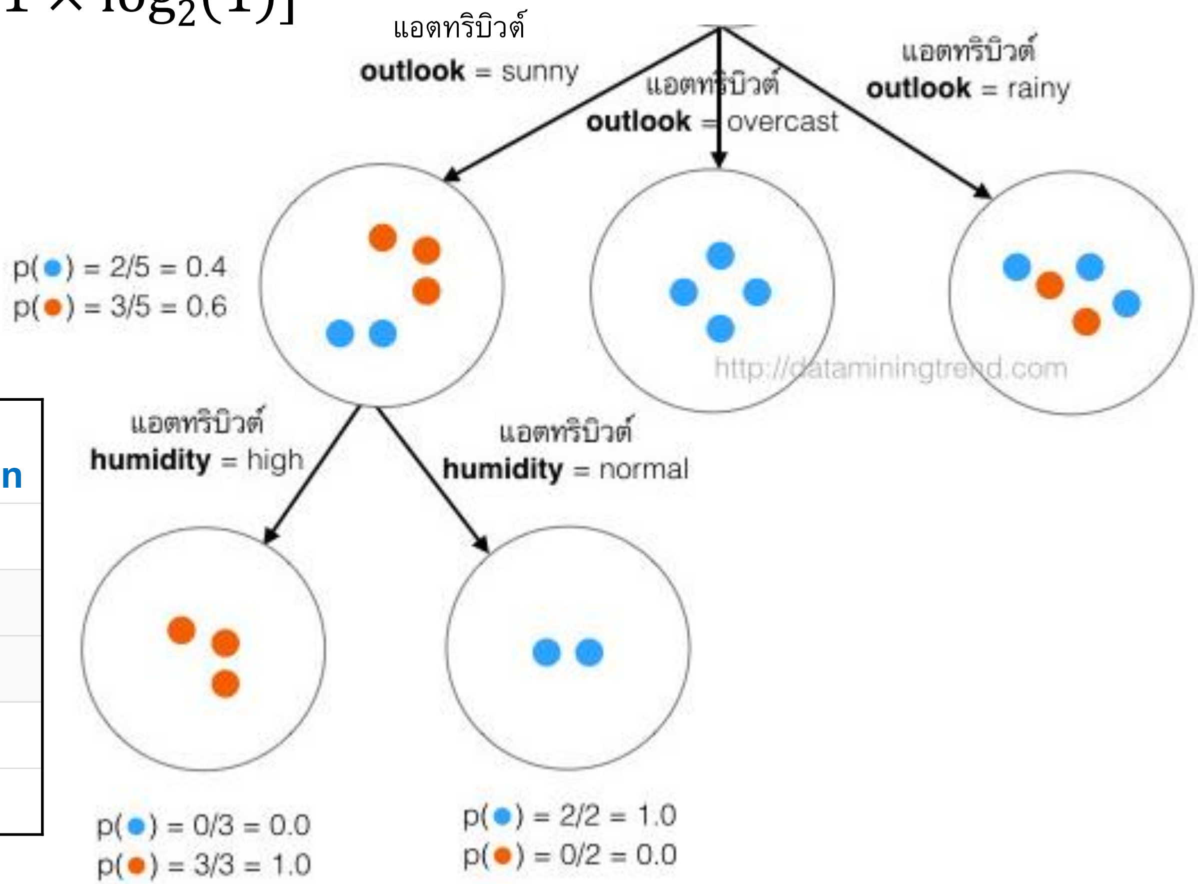
คำนวณค่า Entropy ของ attribute Humidity (เฉพาะที่ Outlook=sunny)

$$\text{Entropy}\ (\textbf{Outlook} = \text{sunny}\ \&\ \textbf{Humidity} = \text{high})$$

$$= -\ [p(yes) \times \log_2 p(yes) + p(no) \times \log_2 p(no)]$$

$$= -[0 \times \log_2(0) + 1 \times \log_2(1)]$$

$$= 0$$



Day	Outlook	Humidity	Play Decision
1	sunny	high	no
2	sunny	high	no
8	sunny	high	no
9	sunny	normal	yes
11	sunny	normal	yes

https://datacubeth.ai/decision-tree/

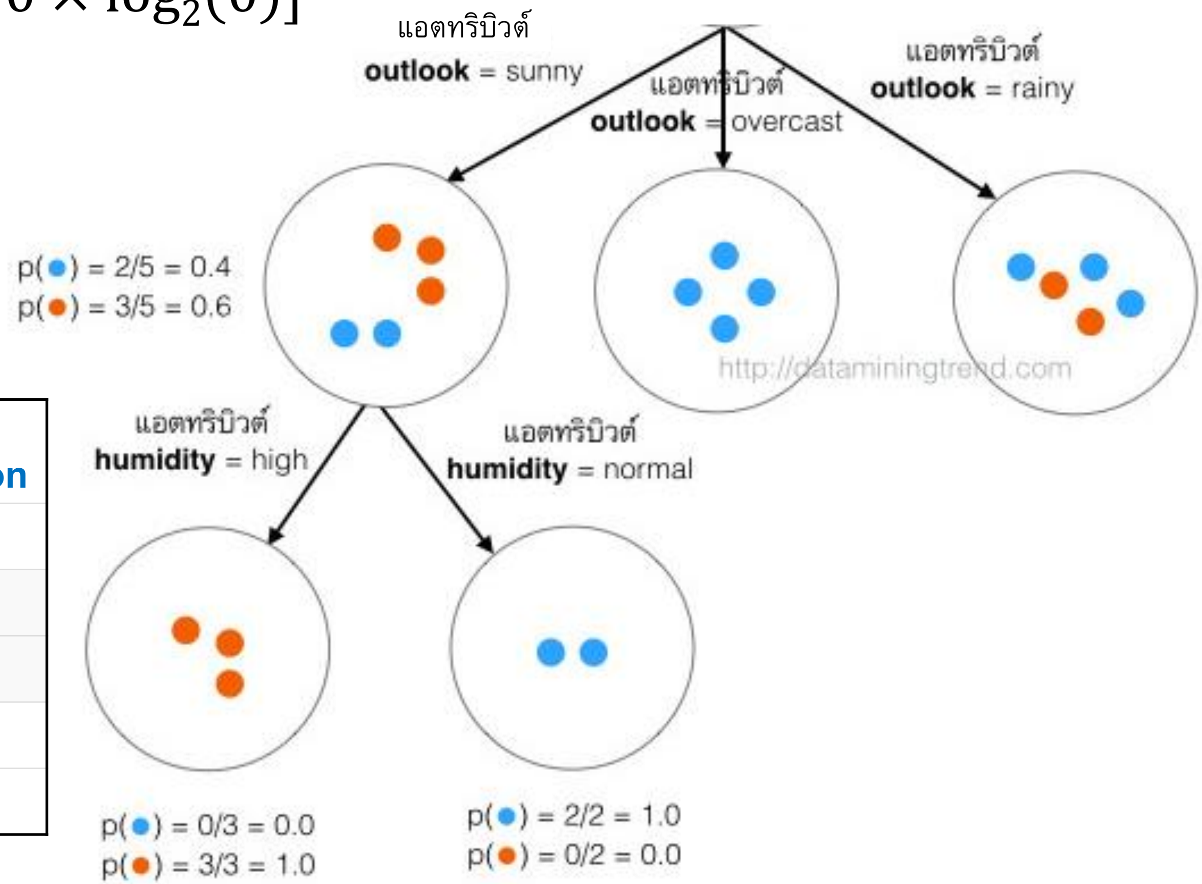
คำนวณค่า Entropy ของ attribute Humidity (เฉพาะที่ Outlook=sunny)

$$\text{Entropy}\ (\textbf{Outlook} = \text{sunny}\ \&\ \textbf{Humidity} = \text{normal})$$

$$= -\ [p(yes) \times \log_2 p(yes) + p(no) \times \log_2 p(no)]$$

$$= -[1 \times \log_2(1) + 0 \times \log_2(0)]$$

$$= 0$$



Day	Outlook	Humidity	Play Decision
1	sunny	high	no
2	sunny	high	no
8	sunny	high	no
9	sunny	normal	yes
11	sunny	normal	yes

https://datacubeth.ai/decision-tree/

คำนวณค่า IG ของ attribute Humidity (เฉพาะที่ Outlook=sunny)

Gain(Outlook=Sunny|Humidity)

Information Gain (Outlook=Sunny|**Humidity**) = Entropy(initial)

– [P(**Outlook**=sunny & **Humidity**=high) × Entropy(**Outlook**=sunny & **Humidity**=high)

+P(**Outlook**=sunny & **Humidity**=normal) × Entropy(**Outlook**=sunny & **Humidity**=normal)]

= 0.97 – [0.6×0 + 0.4x0]

= 0.97

Day	Outlook	Temperature	Humidity	Windy	Play Decision
1	sunny	hot	high	false	no
2	sunny	hot	high	true	no
8	sunny	mild	high	false	no
9	sunny	cool	normal	false	yes
11	sunny	mild	normal	true	yes

Challenge

แสดงวิธีคำนวณค่าที่เหลือ

1- Information Gain (Outlook=Sunny|Temperature)

2- Information Gain (Outlook=Sunny|Windy)

Sunny outlook on decision

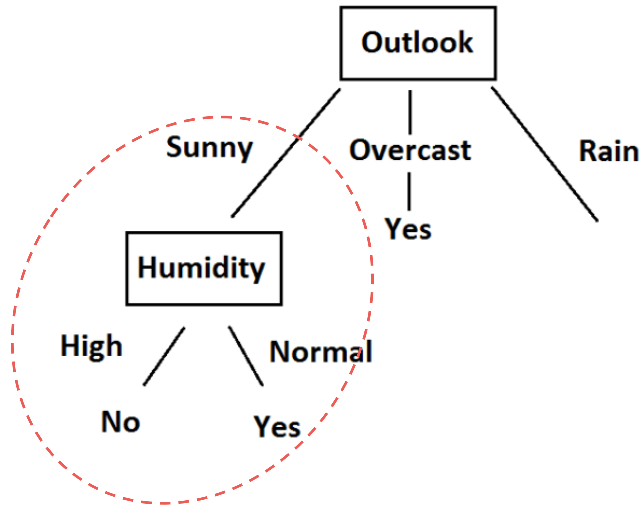
Day	Outlook	Temperature	Humidity	Windy	Play Decision
1	sunny	hot	high	false	no
2	sunny	hot	high	true	no
8	sunny	mild	high	false	no
9	sunny	cool	normal	false	yes
11	sunny	mild	normal	true	yes

There are 5 instances for sunny outlook. Decision would be probably 3/5 percent no, 2/5 percent yes.

1- Gain(Outlook=Sunny|Humidity) = 0.970

2- Gain(Outlook=Sunny|Temperature) = 0.570

3- Gain(Outlook=Sunny|Windy) = 0.019



Now, **humidity** is the decision because it produces the highest score if outlook were sunny

Rain outlook on decision

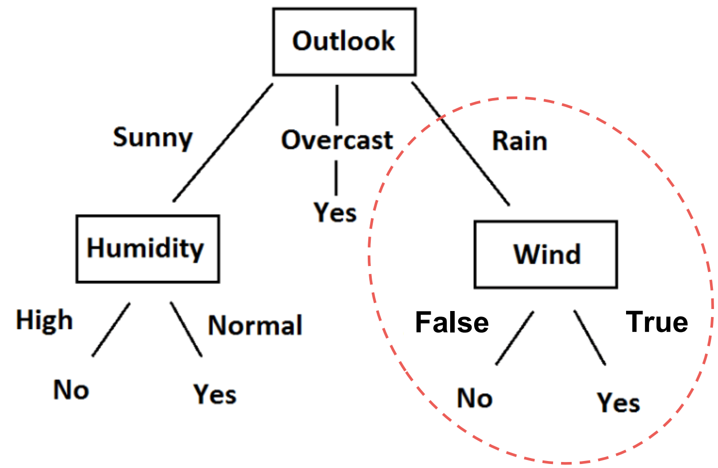
Day	Outlook	Temperature	Humidity	Windy	Play Decision
4	rain	mild	high	false	yes
5	rain	cool	normal	false	yes
6	rain	cool	normal	true	no
10	rain	mild	normal	false	yes
14	rain	mild	high	true	no

There are 5 instances for rain outlook. Decision would be probably 2/5 percent no, 3/5 percent yes.

1- Gain(Outlook=Rain | Temperature)= 0.02

2- Gain(Outlook=Rain | Humidity) = 0.02

3- Gain(Outlook=Rain | Windy) = 0.97

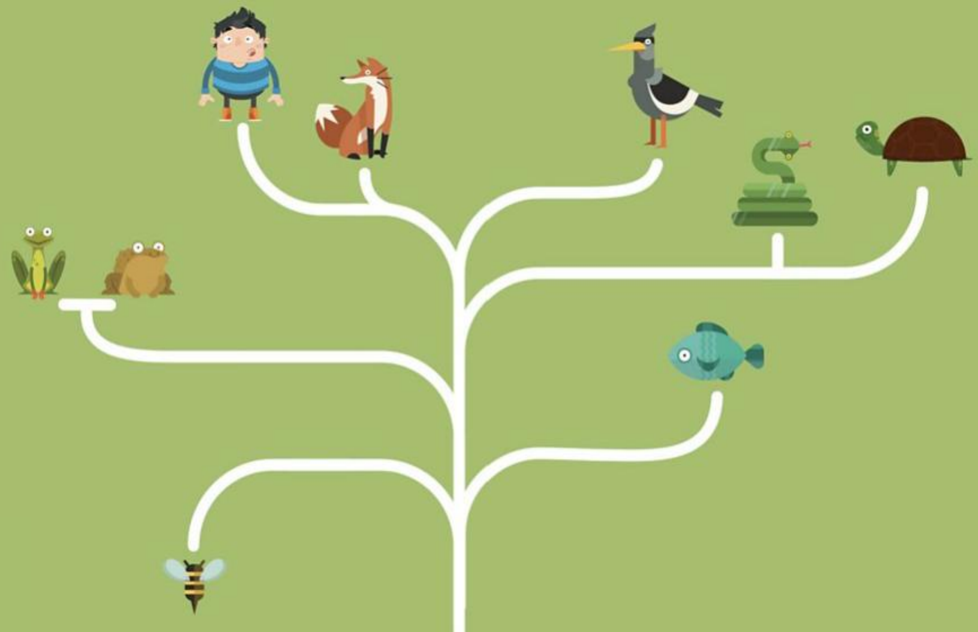


It is revealed that decision will always be yes if outlook=rain and windy=false. While decision will be always no if outlook=rain and windy=true. So, decision tree construction is over. We can use the following rules for decision.

Commonly Used Decision Tree Algorithms

1. ID3 (Iterative Dichotomiser 3)
    - **Main Idea:** The ID3 algorithm uses entropy and information gain to construct decision trees. It iteratively selects the best attribute to split the dataset based on the highest information gain until all data points belong to the same class or no attributes are left.
    - **Limitations:** It cannot handle continuous-valued attributes and may create overfitting trees due to its greedy nature.
2. C4.5 (Classification and Regression Tree)
    - **Main Idea:** C4.5 is an extension of ID3 and addresses some of its limitations. It can handle both categorical and continuous attributes. Instead of information gain, it uses gain ratio as the splitting criterion, which penalizes attributes with many values.
    - **Handling Missing Values:** C4.5 can handle missing values by assigning probabilities to each possible value based on the distribution of the other values in the dataset.
3. CART (Classification and Regression Trees)
4. Random Forest
5. Gradient Boosted Trees

C4.5

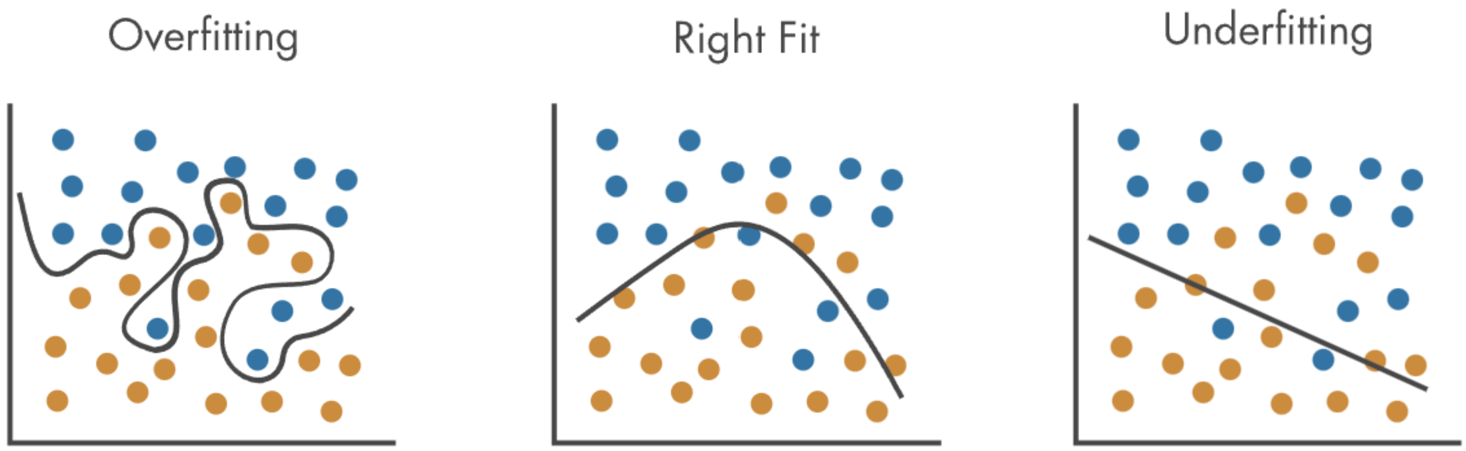


Highly-branching attributes

- Subsets are more likely to be pure if there is a large number of values
  - Information gain is biased towards choosing attributes with a large number of values
  - This may result in overfitting (selection of an attribute that is non-optimal for prediction)

Overfitting

- Overfitting is a machine learning behavior that occurs when the model is so closely aligned to the training data that it does not know how to respond to new data.



Overfitting: Training error is low, but testing error is significantly higher.

Underfitting: Errors are consistently high across training and testing data sets.

Gain ratio

- Gain ratio: a modification of the information gain that reduces its bias on high-branch attributes.
- Gain ratio should be:
  - Large when data is evenly spread.
  - Small when all data belong to one branch.
- Gain ratio takes number and size of branches into account when choosing an attribute.

$$\text{Gain Ratio(S, A)} = \frac{\text{Information Gain(S, A)}}{\text{SplitInfo(S, A)}}$$

$$\text{Split Info(S, A)} = -\sum \left[\frac{|S_v|}{|S|} * \ log_2 \frac{|S_v|}{|S|} \right]$$

Gain ratio

$$\text{Gain Ratio(S, A)} = \frac{\text{Information Gain(S, A)}}{\text{Split Info(S, A)}}$$

$$\text{Split Info(S, A)} = - \sum \left[\frac{|S_v|}{|S|} * \ log_2 \frac{|S_v|}{|S|} \right]$$

where

S is the original dataset.

A is the attribute being used to split the data.

S_v is the subset of S for a specific value of attribute A.

$|S_v|$ is the number of instances in S_v.

|S| is the total number of instances in S.

Day	Outlook	Temperature	Humidity	Windy	Play Decision
1	sunny	85	85	false	no
2	sunny	80	90	true	no
3	overcast	83	78	false	yes
4	rain	70	96	false	yes
5	rain	68	80	false	yes
6	rain	65	70	true	no
7	overcast	64	65	true	yes
8	sunny	72	95	false	no
9	sunny	69	70	false	yes
10	rain	75	80	false	yes
11	sunny	75	70	true	yes
12	overcast	72	90	true	yes
13	overcast	81	75	false	yes
14	rain	71	80	true	no

Wind Attribute

$$\text{IG(parent, child)} = \text{Entropy(parent)} - \sum [\, p(\text{parent}|\text{child}) * \text{Entropy}(\text{parent}|\text{child}) \,]$$

$$= \text{Entropy(parent)} - [\, p(c1) \text{ x } \text{Entropy}(c1) + p(c2) \text{ x } \text{Entropy}(c2) + \ldots \,]$$

$$\text{IG(Decision, } \textbf{Windy}) = \text{Entropy(initial)} - [P(\textbf{Windy}=\text{true}) \times \text{Entropy}(\textbf{Windy}=\text{true})$$

$$+ P(\textbf{Windy}=\text{false}) \times \text{Entropy}(\textbf{Windy}=\text{false}) \,]$$

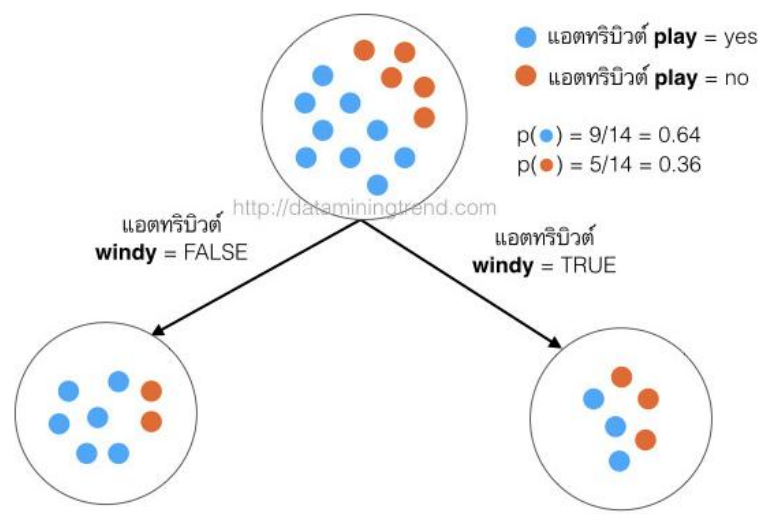
$$= 0.94 - [(6/14)\times 1 + (8/14)\times 0.58] \quad = \quad 0.05$$

$$\text{SplitInfo(Decision, } \textbf{Windy}) = -\, [(6/14)\times \log_2(6/14) + (8/14)\times \log_2(8/14) \,]$$

$$= 0.985$$

$$\text{GainRatio(Decision, } \textbf{Windy}) = 0.05 \,/\, 0.985$$

$$= 0.049$$



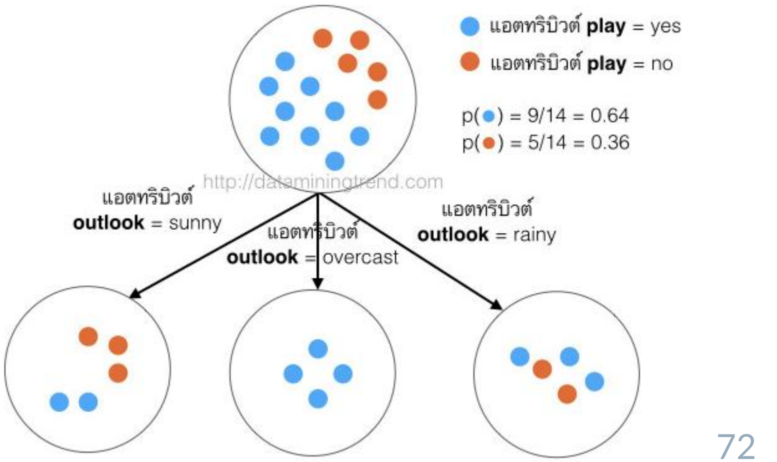
Outlook Attribute

$$IG(parent, child) = Entropy(parent) - \sum [\, p(parent|child) * Entropy(parent|child) \,]$$
$$= Entropy(parent) - [\, p(c1) \times Entropy(c1) + p(c2) \times Entropy(c2) + \ldots \,]$$

$$IG(Decision, \textbf{Outlook}) = Entropy(initial) - [P(\textbf{Outlook}=sunny) \times Entropy(\textbf{Outlook}=sunny)$$
$$+ P(\textbf{Outlook}=overcast) \times Entropy(\textbf{Outlook}=overcast)$$
$$+ P(\textbf{Outlook}=rain) \times Entropy(\textbf{Outlook}=rain)\,]$$
$$= 0.94 - [(5/14)\times 0.97 + (4/14)\times 0 + (5/14)\times 0.97] = 0.246$$

$$SplitInfo(Decision, \textbf{Outlook})$$
$$= -\,[(5/14)\times \log_2(5/14) + (4/14)\times \log_2(4/14) + (5/14)\times \log_2(5/14)\,] = 1.577$$

$$GainRatio(Decision, \textbf{Outlook}) = 0.246 / 1.577$$
$$= 0.155$$



Humidity Attribute

Day	Humidity	Decision
7	65	Yes
6	70	No
9	70	Yes
11	70	Yes
13	75	Yes
3	78	Yes
5	80	Yes
10	80	Yes
14	80	No
1	85	No
2	90	No
12	90	Yes
8	95	No
4	96	Yes

- iterate on all humidity values and separate dataset into two parts as: instances less than or equal to current value, and instances greater than the current value.
- calculate gain ratio for every step.
- the value which maximizes the gain would be the threshold.

$$\text{Entropy(Decision|Humidity}\le 65) = -\,p(\text{No})\,.\,\log_2 p(\text{No}) - p(\text{Yes})\,.\,\log_2 p(\text{Yes})$$
$$= -\,(0/1).\log_2(0/1) - (1/1).\log_2(1/1) = 0$$
$$\text{Entropy(Decision|Humidity}>65) = -\,(5/13).\log_2(5/13) - (8/13).\log_2(8/13) = 0.961$$
$$\text{Gain(Decision, Humidity}<>65) = 0.940 - (1/14)(0) - (13/14)(0.961) = 0.048$$
$$\text{SplitInfo(Decision, Humidity}<>65) = -\,(1/14).\log_2(1/14) - (13/14).\log_2(13/14) = 0.371$$
$$\text{GainRatio(Decision, Humidity}<>65) = 0.126$$

** The statement above refers to that what would branch of decision tree be for less than or equal to 65, and greater than 65. It **does not** refer to that humidity is not equal to 65!*

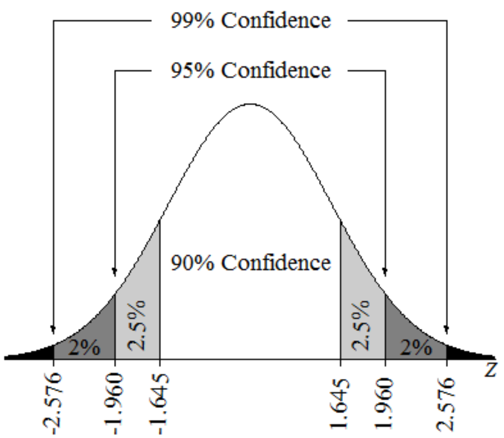
$$\text{Entropy(Decision|Humidity}\le 70) = -\,(1/4).\log_2(1/4) - (3/4).\log_2(3/4) = 0.811$$
$$\text{Entropy(Decision|Humidity}>70) = -\,(4/10).\log_2(4/10) - (6/10).\log_2(6/10) = 0.970$$
$$\text{Gain(Decision, Humidity}<>70) = 0.940 - (4/14)(0.811) - (10/14)(0.970) = 0.014$$
$$\text{SplitInfo(Decision, Humidity}<>70) = -\,(4/14).\log_2(4/14) - (10/14).\log_2(10/14) = 0.863$$
$$\text{GainRatio(Decision, Humidity}<>70) = 0.016$$

Attribute	Gain	Gain Ratio
Wind	0.049	0.049
Outlook	0.246	0.155
Humidity <> 80	0.101	0.107
Temperature <> 83	0.113	0.305

If we will use gain metric, then outlook will be the root node because it has the highest gain value. On the other hand, if we use gain ratio metric, then temperature will be the root node because it has the highest gain ratio value.

Pruning based on confidence intervals

- ID3 splits the tree until all classes of all features in the dataset are used. As the tree gets deeper, overfitting is likely to occur, so pruning is necessary.



- population proportion (p)

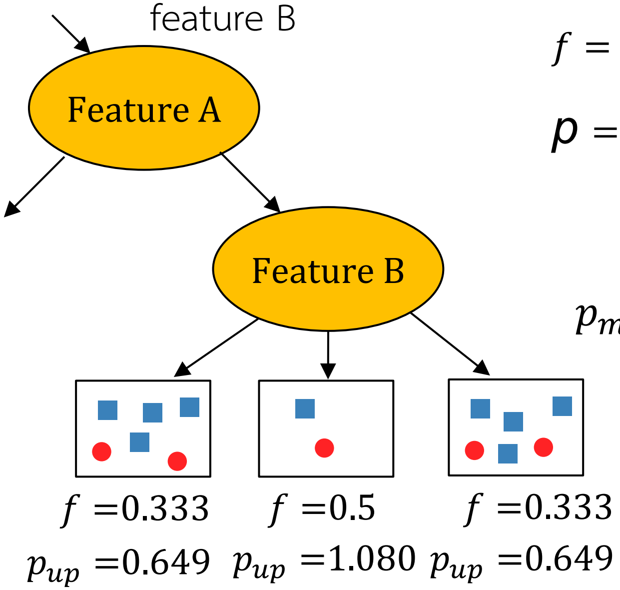
$$p = f \pm z\sqrt{\frac{f(1-f)}{n}} \qquad z=1.645\ (90\%\ \text{CI})$$

- Sample proportion (f) and population proportion (p) before splitting feature B

$f = 5/14 = 0.357$ <-- misclassification rate

$$p = 0.357 \pm 1.645\sqrt{\frac{0.357(1-0.357)}{14}} = 0.147 \sim 0.567$$

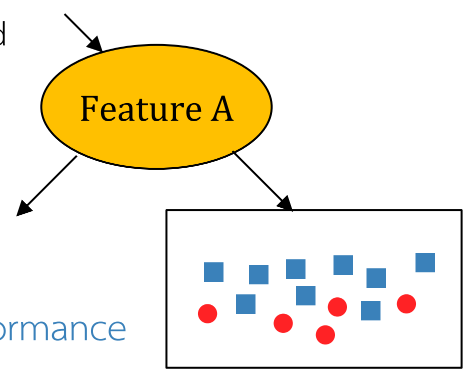
upper bound



$$p_m = 0.649*(6/14) + 1.08*(2/14) + 0.649*(6/14)$$

$= 0.711$ <-- weight avg.

the smaller upper bound, the better performance



It is better not to split feature B

Commonly Used Decision Tree Algorithms

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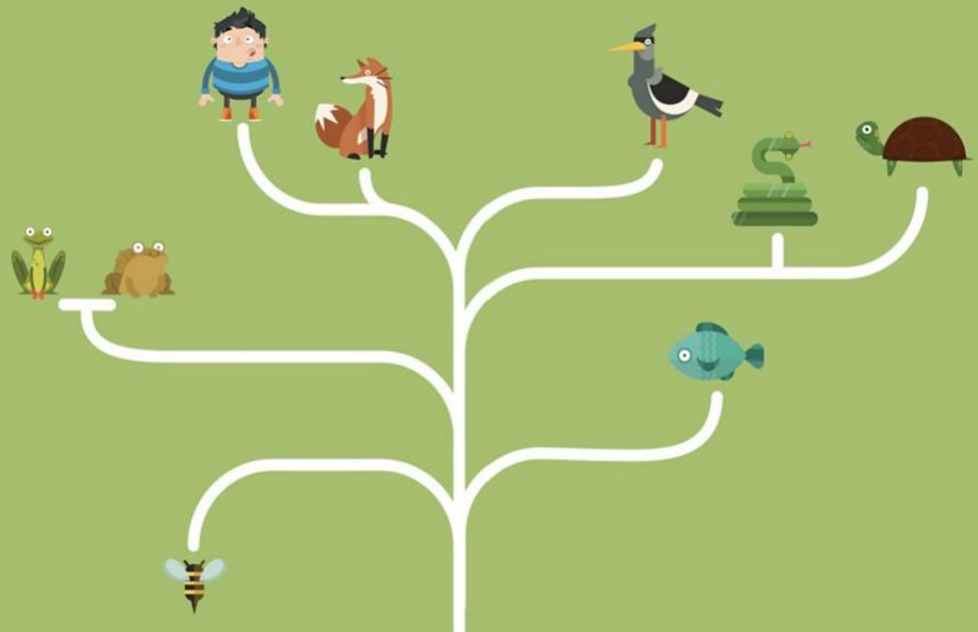
   - **Limitations:** It cannot handle continuous-valued attributes and may create overfitting trees due to its greedy nature.
2. C4.5 (Classification and Regression Tree)
   - **Main Idea:** C4.5 is an extension of ID3 and addresses some of its limitations. It can handle both categorical and continuous attributes. Instead of information gain, it uses gain ratio as the splitting criterion, which penalizes attributes with many values.
   - **Handling Missing Values:** C4.5 can handle missing values by assigning probabilities to each possible value based on the distribution of the other values in the dataset.
3. CART (Classification and Regression Trees)
4. Random Forest
5. Gradient Boosted Trees

CART



Commonly Used Decision Tree Algorithms

1. ID3 (Iterative Dichotomiser 3)
2. C4.5 (Classification and Regression Tree)
3. **CART (Classification and Regression Trees)**
   - **Main Idea:** CART uses Gini impurity to determine the best splits. CART constructs binary trees (each internal node has two children) and recursively splits the dataset into subsets.
4. Random Forest
5. Gradient Boosted Trees

Gini Index (Gini Impurity)

- Gini impurity is a measure of non-homogeneity.
- If a dataset D contains samples from C Classes, Gini index is defined as:

$$\mathrm{Gini(D)} = 1 - \sum_{i=1}^{c} p_i^2$$

where p_i is the probability of a data point belonging to class c in that node.

Example 3

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

คำนวณ Gini ของ root node

$$\text{Gini(D)} = 1 - \sum_{i=1}^{c} p_i^2$$

$$= 1 - [\text{p(sunburned)}^2 + \text{p(none)}^2]$$

$$= 1 - [(3/8)^2 + (5/8)^2] = 0.469$$

If a dataset D splits on S into n subsets D_1, D_2, ..., D_n the Gini index is defined as:

$$\text{Gini}_s(\text{D}) = \frac{D_1}{D}\text{Gini}(\text{D}_1) + \frac{D_2}{D}\text{Gini}(\text{D}_2) + \ldots + \frac{D_n}{D}\text{Gini}(\text{D}_n)$$

weighted avg after split

Split by "Hair"

$$\text{Gini}(D) = 1 - \sum_{i=1}^{c} p_i^2$$

$$= 1 - [p(\text{sunburned})^2 + p(\text{none})^2]$$

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

$$\text{Gini}(\textbf{Hair}=\text{blonde}) = 1 - [(2/4)^2 + (2/4)^2] = 0.5$$

$$\text{Gini}(\textbf{Hair}=\text{brown}) = 1 - [(0/3)^2 + (3/3)^2] = 0$$

$$\text{Gini}(\textbf{Hair}=\text{red}) = 1 - [(1/1)^2 + (0/1)^2] = 0$$

Gini after split (weighted avg):

$$Gini_{Hair} = \frac{4}{8}(0.5) + \frac{3}{8}(0) + \frac{1}{8}(0) = 0.25$$

Split by "Height"

$$\text{Gini(D)} = 1 - \sum_{i=1}^{c} p_i^2$$

$$= 1 - [\text{p(sunburned)}^2 + \text{p(none)}^2]$$

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

$$\text{Gini }(\textbf{Height}=\text{tall}) = 1 - [(0/2)^2 + (2/2)^2] = 0$$

$$\text{Gini }(\textbf{Height}=\text{average}) = 1 - [(2/3)^2 + (1/3)^2] = 0.444$$

$$\text{Gini }(\textbf{Height}=\text{short}) = 1 - [(1/3)^2 + (2/3)^2] = 0.444$$

Gini after split (weighted avg):

$$Gini_{Height} = \frac{3}{8}(0.444) + \frac{2}{8}(0) + \frac{3}{8}(0.444) \approx 0.333$$

Split by "Weight"

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

$$\text{Gini(D)} = 1 - \sum_{i=1}^{c} p_i^2$$

$$= 1 - [p(\text{sunburned})^2 + p(\text{none})^2]$$

Gini (**Weight**=light) = $1 - [(1/2)^2 + (1/2)^2] = 0.5$

Gini (**Weight** =average) = $1 - [(1/3)^2 + (2/3)^2] = 0.444$

Gini (**Weight** =heavy) = $1 - [(1/3)^2 + (2/3)^2] = 0.444$

Gini after split (weighted avg):

$$Gini_{Weight} = \frac{2}{8}(0.5) + \frac{3}{8}(0.444) + \frac{3}{8}(0.444) \approx 0.444$$

Split by "Lotion"

$$\text{Gini(D)} = 1 - \sum_{i=1}^{c} p_i^2$$

$$= 1 - [\text{p(sunburned)}^2 + \text{p(none)}^2]$$

Name	Hair	Height	Weight	Lotion	Result
Sarah	blonde	average	light	no	sunburned
Dana	blonde	tall	average	yes	none
Alex	brown	short	average	yes	none
Annie	blonde	short	average	no	sunburned
Emily	red	average	heavy	no	sunburned
Pete	brown	tall	heavy	no	none
John	brown	average	heavy	no	none
Katie	blonde	short	light	yes	none

Gini (**Lotion**=no) = $1 - [(3/5)^2 + (2/5)^2] = 0.48$

Gini (**Lotion**=yes) = $1 - [(0/3)^2 + (3/3)^2] = 0$

Gini after split (weighted avg):

$$Gini_{Lotion} = \frac{3}{8}(0) + \frac{5}{8}(0.48) = 0.3$$

Gini Index (Gini Impurity)

- เลือก Hair เป็น root node

$$Gini_{Hair} = \frac{4}{8}(0.5) + \frac{3}{8}(0) + \frac{1}{8}(0) = 0.25$$

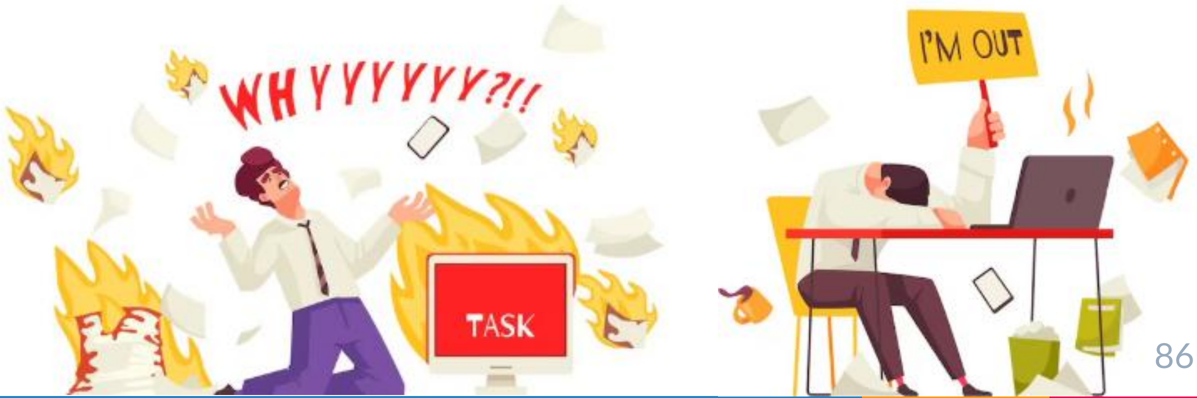
$$Gini_{Height} = \frac{3}{8}(0.444) + \frac{2}{8}(0) + \frac{3}{8}(0.444) \approx 0.333$$

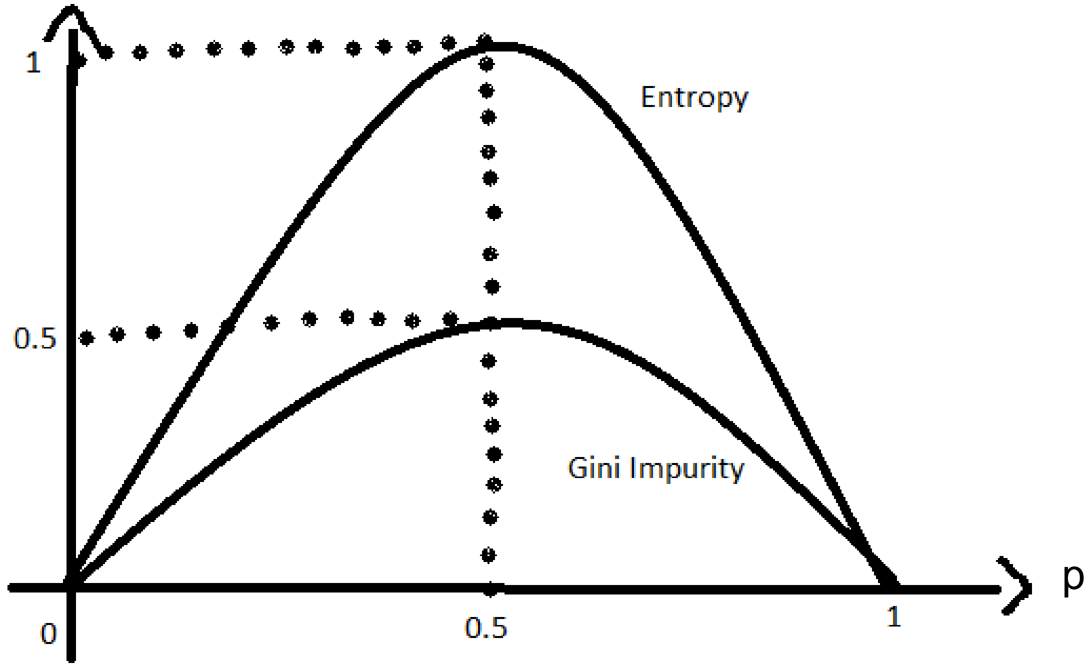
$$Gini_{Weight} = \frac{2}{8}(0.5) + \frac{3}{8}(0.444) + \frac{3}{8}(0.444) \approx 0.444$$

$$Gini_{Lotion} = \frac{3}{8}(0) + \frac{5}{8}(0.48) = 0.3$$

สร้าง Decision Tree

- เนื่องจาก CART เป็น binary tree --> ต้อง split แบบ binary แม้ feature จะมีหลาย category
- วิธีหนึ่งที่นิยมใช้คือ one-vs-rest เช่น :
  - Hair = blonde --> ซ้าย
  - Hair ≠ blonde (i.e., brown, red) --> ขวา





$$E(S) = \sum_{i=1}^{c} - p_i \log_2 p_i$$

$$Gini(E) = 1 - \sum_{j=1}^{c} p_j^2$$

`class sklearn.tree.DecisionTreeClassifier(criterion='gini')`

```
criterion{"gini", "entropy", "log_loss"},
default="gini"
```

Imbalance Dataset

นิยาม:

- ใน Binary Classification - Class หนึ่งมีจำนวน 90% และอีก Class เพียง 10%
- ใน Multi-classes Classification - Class หนึ่ง มีจำนวน Samples มากกว่าหรือน้อยกว่า Class อื่น ๆ อย่างมาก

ปัญหาที่เกิดขึ้น:

- ML Algorithms ส่วนใหญ่มักจะโน้มเอียง (Bias) ไปทาง Class ส่วนใหญ่ ซึ่งนำไปสู่ประสิทธิภาพที่ไม่ดีใน Class ส่วนน้อย
- Model อาจมีความแม่นยำโดยรวมสูง แต่ประสิทธิภาพไม่ดีในการทำงานกับ Class ส่วนน้อย
- Standard metric ที่ใช้ในการประเมิน Model เช่น Accuracy อาจทำให้ตีความหมายผิดได้

ตัวอย่างในโลกแห่งความจริง:

- การตรวจจับการฉ้อโกง: ธุรกรรมส่วนใหญ่ถูกต้องตามกฎหมาย โดยที่การฉ้อโกงเกิดขึ้นน้อย
- การวินิจฉัยทางการแพทย์: กรณีของโรคมักมีน้อยกว่ากรณีที่สุขภาพดี
- การตรวจจับอีเมลสแปม: อีเมลสแปมมักจะพบน้อยกว่าอีเมลที่ถูกต้อง

Cohen's Kappa

- A statistic that measures inter-annotator agreement.
- a score that expresses the level of agreement between two annotators on a classification problem. It is defined as:

$$\kappa = (p_o - p_e)/(1 - p_e)$$

where

p_o is the empirical probability of agreement on the label assigned to any sample (สัดส่วนความแม่นยำที่เกิดขึ้นจริง - the observed agreement ratio)

p_e is the expected agreement when both annotators assign labels randomly. (สัดส่วนความแม่นยำที่เกิดขึ้นแบบสุ่ม)

p_e is estimated using a per-annotator empirical prior over the class labels

https://scikit-learn.org/stable/modules/generated/sklearn.metrics.cohen_kappa_score.html

Cohen's Kappa

	Model Classification	Human Judgement
Email 1	YES	YES
Email 2	YES	NO
Email 3	YES	YES
Email 4	YES	YES
Email 5	NO	YES
Email 6	NO	NO
..	--	--
Email 100	YES	YES

		Human Judgement		
		Yes	No	
Model Classification	Yes	34	16	**50**
	No	12	38	**50**
		46	**54**	

Cohen's Kappa

		Human Judgement		
		Yes	No	
Model Classification	Yes	34	16	**50**
	No	12	38	**50**
		46	**54**	

Po : Relative observed agreement among raters

Po = (Both said Yes + Both said No) / (Total Rating)

Po = (34 + 38) / (100) = 0.72

The probability *Po* is 72% or 0.72

K = (Po - Pe) / (1 - Pe)

= (0.72 – 0.5) / (1 – 0.5) = 0.44

The Kappa value is 0.4

Pe : Hypothetical probability of chance agreement (the probabilities of each observer randomly selecting each category)

P("Yes") = [(34+16)/100] * [(34+12)/100] = 0.23

P("No") = [(12+38)/100] * [(16+38)/100] = 0.27

Pe = 0.23 + 0.27 = 0.5

Cohen’s Kappa

K ≤ 0: Indicates no agreement or agreement worse than chance.

0.01 – 0.20: Slight agreement.

0.21 – 0.40: Fair agreement.

0.41 – 0.60: Moderate agreement.

0.61 – 0.80: Substantial agreement.

0.81 – 1.00: Almost perfect agreement.

Cohen’s Kappa Example

สรุป

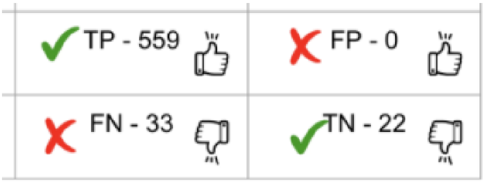
- Cohen’s Kappa เป็นวิธีวัดความแม่นยำของโมเดล classification ที่ “ปรับค่า accuracy ให้คำนึงถึงความบังเอิญในการทำนาย” โดยเฉพาะในกรณีที่ class ไม่สมดุล หรือมีความเป็นไปได้ที่จะทายถูกโดยไม่ตั้งใจ
- Pe = ความแม่นที่คาดว่าจะเกิดจากการเดาสุ่มตามสัดส่วน class
- ถ้า accuracy ของเราดีกว่า Pe มาก → Kappa จะสูง
- ถ้า accuracy พอ ๆ กับ Pe → Kappa จะใกล้ 0

https://scikit-learn.org/stable/modules/generated/sklearn.metrics.cohen_kappa_score.html

Matthews Correlation Coefficient (MCC)

- MCC assesses the correlation between predicted and actual binary outcomes.
- Unlike accuracy, MCC is not significantly affected by class imbalance. It provides a more reliable measure of performance when one class significantly outnumbers the other.

$$\mathrm{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$



- Interpretation:
  - +1: Perfect prediction.
  - 0: Random prediction.
  - -1: Complete disagreement between prediction and actual values.
- MCC is recommended when dealing with imbalanced datasets or when both positive and negative classes are equally important. It is a robust metric for evaluating binary classification models

Matthews Correlation Coefficient (MCC)

- Key Differences Summarized:

Feature	Cohen's Kappa	Matthews Correlation Coefficient (MCC)
Main Use	Inter-rater reliability, agreement assessment	Binary classification performance evaluation
Class Imbalance	Sensitive	Robust
Interpretation	Agreement beyond chance	Correlation between prediction and reality
Range	-1 to +1 (sometimes undefined)	-1 to +1
Best Use	When agreement between raters or classifiers is the primary focus	When evaluating classifier performance on imbalanced datasets or when a balanced assessment is desired

The Matthews Correlation Coefficient (MCC) is More Informative Than Cohen's Kappa and Brier Score in Binary Classification Assessment https://ieeexplore.ieee.org/document/9440903